Searching Central Difference Convolutional Networks for Face Anti-Spoofing

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Abstract

Face anti-spoofing (FAS) plays a vital role in face recognition systems. Most state-of-the-art FAS methods 1) rely on stacked convolutions and expert-designed network, which is weak in describing detailed fine-grained information and easily being ineffective when the environment varies (e.g., different illumination), and 2) prefer to use long sequence as input to extract dynamic features, making them difficult to deploy into scenarios which need quick response. Here we propose a novel frame level FAS method based on Central Difference Convolution (CDC), which is able to capture intrinsic detailed patterns via aggregating both intensity and gradient information. A network built with CDC, called the Central Difference Convolutional Network (CDCN), is able to provide more robust modeling capacity than its counterpart built with vanilla convolution. Furthermore, over a specifically designed CDC search space, Neural Architecture Search (NAS) is utilized to discover a more powerful network structure (CDCN++), which can be assembled with Multiscale Attention Fusion Module (MAFM) for further boosting performance. Comprehensive experiments are performed on six benchmark datasets to show that 1) the proposed method not only achieves superior performance on intra-dataset testing (especially 0.2% ACER in Protocol-1 of OULU-NPU dataset), 2) it also generalizes well on cross-dataset testing (particularly 6.5% HTER from CASIA-MFSD to Replay-Attack datasets). The codes are available at https://github.com/ZitongYu/CDCN.

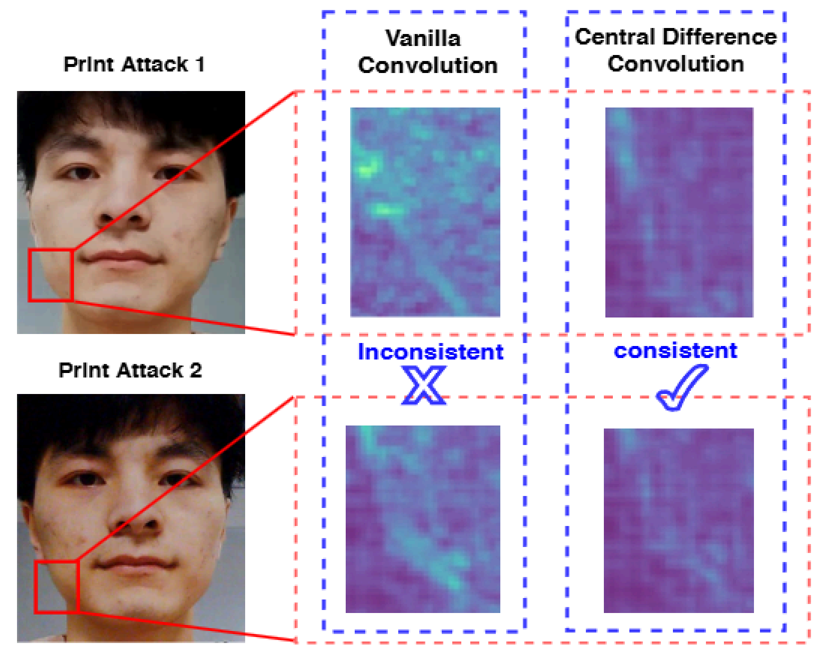

Figure 1. Feature response of vanilla convolution (VanillaConv) and central difference convolution (CDC) for spoofing faces in shifted domains (illumination & input camera). VanillaConv fails to capture the consistent spoofing pattern while CDC is able to extract the invariant detailed spoofing features, e.g., **lattice artifacts**.

1. Introduction

Face recognition has been widely used in many interactive artificial intelligence systems for its convenience. However, vulnerability to presentation attacks (PA) curtail its reliable deployment. Merely presenting printed images or videos to the biometric sensor could fool face recognition systems. Typical examples of presentation attacks are print, video replay, and 3D masks. For the reliable use of face recognition systems, face anti-spoofing (FAS) methods are important to detect such presentation attacks.

In recent years, several hand-crafted features based [7, 8, 15, 29, 45, 44] and deep learning based [49, 64, 36, 26, 62, 4, 19, 20] methods have been proposed for presentation attack detection (PAD). On one hand, the classical hand-crafted descriptors (e.g., local binary pattern (LBP) [7]) leverage local relationship among the neighbours as the discriminative features, which is robust for describing the detailed invariant information (e.g., color texture, moiré pattern and noise artifacts) between the living and spoofing faces. On the other hand, due to the stacked convolution operations with nonlinear activation, the convolutional neural networks (CNN) hold strong representation abilities to distinguish the bona fide and PA. However, CNN based methods focus on the deeper semantic features, which are weak in describing detailed fine-grained information between liv-

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搜索中心差分卷积网络用于人脸防伪

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摘要

人脸防欺骗（FAS）在人脸识别系统中发挥着至关重要的作用。大多数最先进的 FAS 方法 1）依赖于堆叠卷积和专家设计的网络，这在描述详细的细粒度信息方面较弱，并且在环境变化时（例如，不同的光照条件）容易失效，以及 2）倾向于使用长序列作为输入来提取动态特征，这使得它们难以部署到需要快速响应的场景中。在这里，我们提出了一种基于中心差分卷积（CDC）的帧级 FAS 方法，该方法能够通过聚合强度和梯度信息来捕获内在的详细模式。一个使用 CDC 构建的网络，称为中心差分卷积网络（CDCN），能够提供比使用普通卷积构建的对应网络更强的建模能力。此外，在一个专门设计的 CDC 搜索空间中，利用神经架构搜索（NAS）发现了一个更强大的网络结构（CDCN++），它可以与多尺度注意力融合模块（MAFM）结合使用，以进一步提升性能。 在六个基准数据集上进行了全面的实验，以证明 1) 所提出的方法不仅在数据集内测试中取得了优异的性能（特别是在 OULU-NPU 数据集的 Protocol1 中达到了 0.2%的 ACER），2) 它在跨数据集测试中也表现良好（特别是在从 CASIAMFSD 到 Replay-Attack 数据集的测试中达到了 6.5%的 HTER）。代码可在 https://github.com/ZitongYu/CDCN 获取。

1. 引言 人脸识别因其便捷性已被广泛应用于许多交互式人工智能系统中。然而，其对呈现攻击（PA）的脆弱性限制了其可靠部署。仅仅向生物识别传感器展示打印的图像或视频就可能欺骗人脸识别

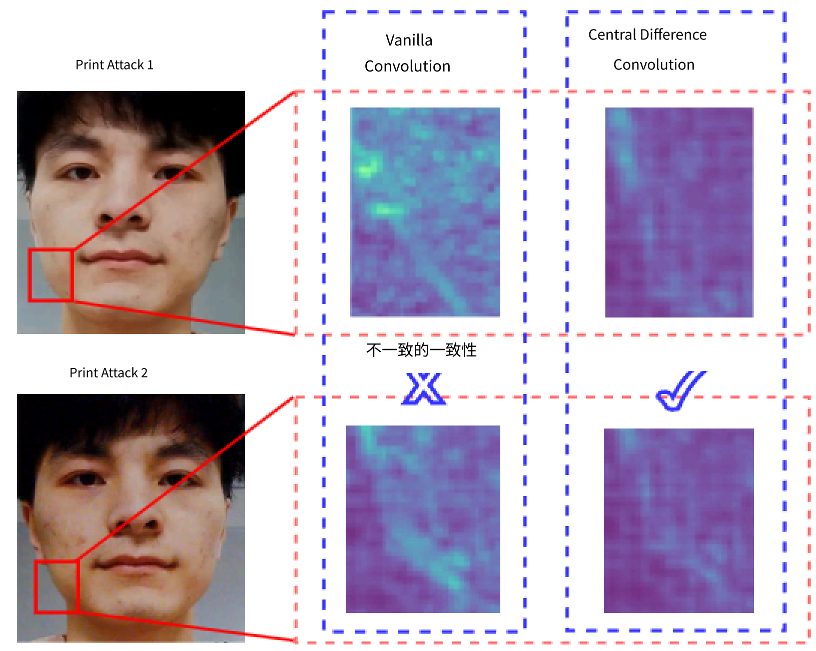

图 1. 普通卷积（VanillaConv）和中心差分卷积（CDC）在移位域（光照与输入相机）中伪造人脸的特征响应。VanillaConv 无法捕捉到一致性的伪造模式，而 CDC 能够提取不变性的详细伪造特征，例如晶格伪影。

系统。典型的演示攻击例子包括打印、视频重放和 3D 面具。为了可靠地使用人脸识别系统，人脸防伪（FAS）方法对于检测此类演示攻击非常重要。

近年来，针对演示攻击检测（PAD），基于手工特征[7, 8, 15, 29, 45, 44]和基于深度学习[49, 64, 36, 26, 62, 4, 19, 20]的方法已被提出。一方面，经典的手工特征描述符（例如局部二值模式（LBP）[7]）利用邻居之间的局部关系作为判别特征，这对于描述真实人脸和欺骗人脸之间详细的不变信息（例如颜色纹理、摩尔纹和噪声伪影）是稳健的。另一方面，由于卷积操作与非线性激活的堆叠，卷积神经网络（CNN）具有强大的表征能力来区分真实人脸和演示攻击。然而，基于 CNN 的方法侧重于更深层的语义特征，这些特征在描述真实人脸和演示攻击之间详细的细粒度信息方面较弱。

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ing and spoofing faces and easily being ineffective when the environment varies (e.g., different light illumination). **How to integrate local descriptors with convolution operation for robust feature representation is worth exploring.**

Most recent deep learning based FAS methods are usually built upon image classification task based backbones [61, 62, 20], such as VGG [54], ResNet [22] and DenseNet [23]. The networks are usually supervised by binary cross-entropy loss, which easily learns the arbitrary patterns such as screen bezel instead of the nature of spoofing patterns. In order to solve this issue, several depth supervised FAS methods [4, 36], which utilize pseudo depth map label as auxiliary supervised signal, have been developed. However, all these network architectures are carefully designed by human experts, which might not be optimal for FAS task. Hence, **to automatically discover best-suited networks for FAS task with auxiliary depth supervision should be considered.**

Most existing state-of-the-art FAS methods [36, 56, 62, 32] need multiple frames as input to extract dynamic spatio-temporal features (e.g., motion [36, 56] and rPPG [62, 32]) for PAD. However, long video sequence may not be suitable for specific deployment conditions where the decision needs to be made quickly. Hence, frame level PAD approaches are advantageous from the usability point of view despite inferior performance compared with video level methods. **To design high-performing frame level methods is crucial for real-world FAS applications.**

Motivated by the discussions above, we propose a novel convolution operator called Central Difference Convolution (CDC), which is good at describing fine-grained invariant information. As shown in Fig. 1, CDC is more likely to extract intrinsic spoofing patterns (e.g., lattice artifacts) than vanilla convolution in diverse environments. Furthermore, over a specifically designed CDC search space, Neural Architecture Search (NAS) is utilized to discover the excellent frame level networks for depth supervised face anti-spoofing task. Our contributions include:

- We design a novel convolution operator called Central Difference Convolution (CDC), which is suitable for FAS task due to its remarkable representation ability for invariant fine-grained features in diverse environments. Without introducing any extra parameters, CDC can replace the vanilla convolution and plug and play in existing neural networks to form Central Difference Convolutional Networks (CDCN) with more robust modeling capacity.
- We propose CDCN++, an extended version of CDCN, consisting of the searched backbone network and Multiscale Attention Fusion Module (MAFM) for aggregating the multi-level CDC features effectively.
- To our best knowledge, this is the first approach that searches neural architectures for FAS task. Different from the previous classification task based NAS supervised by softmax loss, we search the well-suited frame level networks for depth supervised FAS task over a specifically designed CDC search space.
- Our proposed method achieves state-of-the-art performance on all six benchmark datasets with both intra- as well as cross-dataset testing.

2. Related Work

Face Anti-Spoofing. Traditional face anti-spoofing methods usually extract hand-crafted features from the facial images to capture the spoofing patterns. Several classical local descriptors such as LBP [7, 15], SIFT [44], SURF [9], HOG [29] and DoG [45] are utilized to extract frame level features while video level methods usually capture dynamic clues like dynamic texture [28], micro-motion [53] and eye blinking [41]. More recently, a few deep learning based methods are proposed for both frame level and video level face anti-spoofing. For frame level methods [30, 43, 20, 26], pre-trained deep CNN models are fine-tuned to extract features in a binary-classification setting. In contrast, auxiliary depth supervised FAS methods [4, 36] are introduced to learn more detailed information effectively. On the other hand, several video level CNN methods are presented to exploit the dynamic spatio-temporal [56, 62, 33] or rPPG [31, 36, 32] features for PAD. Despite achieving state-of-the-art performance, video level deep learning based methods need long sequence as input. In addition, compared with traditional descriptors, CNN overfits easily and is hard to generalize well on unseen scenes.

Convolution Operators. The convolution operator is commonly used in extracting basic visual features in deep learning framework. Recently extensions to the vanilla convolution operator have been proposed. In one direction, classical local descriptors (e.g., LBP [2] and Gabor filters [25]) are considered into convolution design. Representative works include Local Binary Convolution [27] and Gabor Convolution [38], which is proposed for saving computational cost and enhancing the resistance to the spatial changes, respectively. Another direction is to modify the spatial scope for aggregation. Two related works are dialated convolution [63] and deformable convolution [14]. However, these convolution operators may not be suitable for FAS task because of the limited representation capacity for invariant fine-grained features.

Neural Architecture Search. Our work is motivated by recent researches on NAS [11, 17, 35, 47, 68, 69, 60], while we focus on searching for a depth supervised model with high performance instead of a binary classification model for face anti-spoofing task. There are three main categories of existing NAS methods: 1) Reinforcement learning based [68, 69], 2) Evolution algorithm based [51, 52], and 3) Gradient based [35, 60, 12]. Most of NAS approaches search

颜色纹理识别和伪造面部，在环境变化时（例如不同的光照条件）容易失效。如何将局部描述符与卷积操作结合，以实现鲁棒的特征表示，是值得探索的课题。

目前基于深度学习的 FAS 方法通常建立在图像分类任务的基础骨干上[61, 62, 20]，例如 VGG[54]、ResNet[22]和DenseNet[23]。这些网络通常由二元交叉熵损失进行监督，这容易学习到任意模式（如屏幕边框）而不是伪造模式的本质。为了解决这个问题，已经开发出几种深度监督的 FAS 方法[4, 36]，这些方法利用伪深度图标签作为辅助监督信号。然而，所有这些网络架构都是由人类专家精心设计的，可能并不适合 FAS 任务。因此，应该考虑自动发现最适合 FAS 任务的、带有辅助深度监督的网络。

现有的最先进人脸欺骗检测（FAS）方法[36, 56, 62, 32]需要多帧输入来提取动态时空特征（例如，运动[36, 56]和rPPG[62, 32]）用于伪影检测（PAD）。然而，长视频序列可能不适用于需要快速做出决策的特定部署条件。因此，尽管与视频级方法相比性能较差，但帧级 PAD 方法从可用性角度来看具有优势。设计高性能的帧级方法对于实际人脸欺骗检测应用至关重要。

受上述讨论的启发，我们提出了一种名为中心差分卷积（CDC）的新型卷积算子，该算子擅长描述细粒度不变信息。如图 1 所示，相比于普通卷积，CDC 更倾向于在多种环境中提取内在欺骗模式（例如，晶格伪影）。此外，在专门设计的 CDC 搜索空间中，利用神经架构搜索（NAS）来发现适用于深度监督人脸欺骗检测任务的优秀帧级网络。我们的贡献包括：

- 我们设计了一种名为中心差分卷积（CDC）的新型卷积算子，它适用于 FAS 任务，因为它在多种环境中对不变细粒度特征具有卓越的表征能力。无需引入任何额外参数，CDC 可以替代传统卷积，并即插即用地集成到现有神经网络中，形成具有更强建模能力的中心差分卷积网络（CDCN）。

- 我们提出了 CDCN++，即 CDCN 的扩展版本，它由搜索到的骨干网络和多尺度注意力融合模块（MAFM）组成，用于有效地聚合多级 CDC 特征。

- 据我们所知，这是首个为 FAS 任务搜索神经架构的方法。与先前基于 softmax 损失的分类任务监督搜索不同，我们在专门设计的 CDC 搜索空间中搜索最适合的框架级网络，用于深度监督的 FAS 任务。

- 我们提出的方法在所有六个基准数据集上均实现了最先进的性能，包括数据集内部和跨数据集测试。

2. 相关工作

人脸防欺骗。传统的人脸防欺骗方法通常从人脸图像中提取手工特征来捕捉欺骗模式。几种经典的局部描述符，如 LBP [7, 15]、SIFT [44]、SURF [9]、HOG [29]和 DoG [45]，被用于提取帧级特征，而视频级方法通常捕捉动态线索，如动态纹理 [28]、微运动 [53]和眨眼 [41]。最近，一些基于深度学习的方法被提出用于帧级和视频级的人脸防欺骗。对于帧级方法 [30, 43, 20, 26]，预训练的深度 CNN 模型在二元分类设置中被微调以提取特征。相比之下，辅助深度监督 FAS 方法 [4, 36]被引入以有效学习更详细的信息。另一方面，一些视频级 CNN 方法被提出以利用动态时空 [56, 62, 33] 或 rPPG [31, 36, 32] 特征进行 PAD。尽管取得了最先进的性能，视频级基于深度学习的方法需要长序列作为输入。此外，与传统描述符相比，CNN 容易过拟合，并且难以泛化到未见过的场景。

卷积算子。卷积算子在深度学习框架中常用于提取基本视觉特征。最近提出了对原版卷积算子的扩展。在一个方向上，将经典局部描述符（例如 LBP [2] 和 Gabor 滤波器 [25]）考虑进卷积设计中。代表性工作包括局部二值卷积 [27] 和 Gabor 卷积 [38]，分别提出用于节省计算成本和增强对空间变化的抵抗能力。另一个方向是修改聚合的空间范围。相关的工作包括扩张卷积 [63] 和可变形卷积 [14]。然而，由于对不变细粒度特征的表示能力有限，这些卷积算子可能不适合 FAS 任务。

神经架构搜索。我们的工作受近期关于 NAS 的研究[11, 17, 35, 47, 68, 69, 60]的启发，而我们专注于搜索高性能的深度监督模型，而不是用于人脸防伪任务的二分类模型。现有的 NAS 方法主要分为三类：1) 基于强化学习[68, 69]，2) 基于进化算法[51, 52]，和 3) 基于梯度[35, 60, 12]。大多数 NAS 方法搜索

networks on a small proxy task and transfer the found architecture to another large target task. For the perspective of computer vision applications, NAS has been developed for face recognition [67], action recognition [46], person ReID [50], object detection [21] and segmentation [65] tasks. To the best of our knowledge, no NAS based method has ever been proposed for face anti-spoofing task.

In order to overcome the above-mentioned drawbacks and fill in the blank, we search the frame level CNN over a specially designed search space with the new proposed convolution operator for depth-supervised FAS task.

3. Methodology

In this section, we will first introduce our Central Difference Convolution in Section 3.1, then introduce the **C**entral **D**ifference **C**onvolutional **N**etworks (CDCN) for face anti-spoofing in Section 3.2, and at last present the searched networks with attention mechanism (CDCN++) in Section 3.3.

3.1. Central Difference Convolution

In modern deep learning frameworks, the feature maps and convolution are represented in 3D shape (2D spatial domain and extra channel dimension). As the convolution operation remains the same across the channel dimension, for simplicity, in this subsection the convolutions are described in 2D while extension to 3D is straightforward.

Vanilla Convolution. As 2D spatial convolution is the basic operation in CNN for vision tasks, here we denote it as vanilla convolution and review it shortly first. There are two main steps in the 2D convolution: 1) *sampling* local receptive field region $\mathcal{R}$ over the input feature map x; 2) *aggregation* of sampled values via weighted summation. Hence, the output feature map y can be formulated as

$$y(p_0) = \sum_{p_n \in \mathcal{R}} w(p_n) \cdot x(p_0 + p_n), \tag{1}$$

where p_0 denotes current location on both input and output feature maps while p_n enumerates the locations in $\mathcal{R}$. For instance, local receptive field region for convolution operation with 3×3 kernel and dilation 1 is $\mathcal{R} = \{(-1,-1), (-1,0), \cdots, (0,1), (1,1)\}$.

Vanilla Convolution Meets Central Difference. Inspired by the famous local binary pattern (LBP) [7] which describes local relations in a binary central difference way, we also introduce central difference into vanilla convolution to enhance its representation and generalization capacity. Similarly, central difference convolution also consists of two steps, i.e., *sampling* and *aggregation*. The sampling step is similar to that in vanilla convolution while the aggregation step is different: as illustrated in Fig. 2, central difference convolution prefers to aggregate the center-oriented gradient of sampled values. Eq. (1) becomes

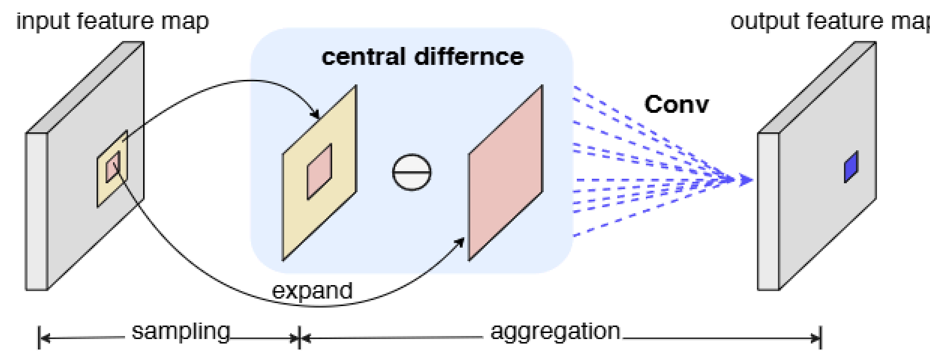

Figure 2. Central difference convolution.

$$y(p_0) = \sum_{p_n \in \mathcal{R}} w(p_n) \cdot (x(p_0 + p_n) - x(p_0)). \tag{2}$$

When $p_n = (0,0)$, the gradient value always equals to zero with respect to the central location p_0 itself.

For face anti-spoofing task, both the intensity-level semantic information and gradient-level detailed message are crucial for distinguishing the living and spoofing faces, which indicates that combining vanilla convolution with central difference convolution might be a feasible manner to provide more robust modeling capacity. Therefore we generalize central difference convolution as

$$\begin{aligned} y(p_0) &= \theta \cdot \underbrace{\sum_{p_n \in \mathcal{R}} w(p_n) \cdot (x(p_0 + p_n) - x(p_0))}_{\text{central difference convolution}} \\ &\quad + (1-\theta) \cdot \underbrace{\sum_{p_n \in \mathcal{R}} w(p_n) \cdot x(p_0 + p_n)}_{\text{vanilla convolution}}, \end{aligned} \tag{3}$$

where hyperparameter $\theta \in [0,1]$ tradeoffs the contribution between intensity-level and gradient-level information. The higher value of θ means the more importance of central difference gradient information. We will henceforth refer to this generalized **C**entral **D**ifference **C**onvolution as **CDC**, which should be easy to identify according to its context.

Implementation for CDC. In order to efficiently implement CDC in modern deep learning framework, we decompose and merge Eq. (3) into the vanilla convolution with additional central difference term

$$y(p_0) = \underbrace{\sum_{p_n \in \mathcal{R}} w(p_n) \cdot x(p_0 + p_n)}_{\text{vanilla convolution}} + \underbrace{\theta \cdot (-x(p_0) \cdot \sum_{p_n \in \mathcal{R}} w(p_n))}_{\text{central difference term}}. \tag{4}$$

According to the Eq. (4), CDC can be easily implemented by a few lines of code in PyTorch[42] and TensorFlow[1]. The derivation of Eq. (4) and codes based on Pytorch are shown in ***Appendix A***.

Relation to Prior Work. Here we discuss the relations between CDC and vanilla convolution, local binary convolution[27] and gabor convolution[38], which share

现有的 NAS 方法主要分为三类：1) 基于强化学习[68]，通过在小型代理任务上搜索网络架构，并将找到的架构迁移到另一个大型目标任务。从计算机视觉应用的角度来看，NAS 已被开发用于人脸识别[67]、动作识别[46]、行人重识别[50]、目标检测[21]和分割[65]任务。据我们所知，还没有基于 NAS 的方法被提出用于人脸防伪任务。

为了克服上述缺点并填补空白，我们使用专门设计的搜索空间，并采用新提出的卷积算子，搜索适用于深度监督人脸防伪任务的 CNN 框架级模型。

3. 方法论 在本节中，我们将首先在 3.1 节介绍我们的中心差分卷积，然后在 3.2 节介绍用于人脸防伪的中心差分卷积网络（CDCN），最后在 3.3 节展示带有注意力机制的被搜索网络（CDCN++）。

3.1. 中心差分卷积 在现代深度学习框架中，特征图和卷积以 3D 形状表示（2D 空间域和额外的通道维度）。由于卷积操作在通道维度上保持不变，为了简化，在本小节中，卷积以 2D 形式描述，而扩展到 3D 是直接的。

普通卷积。由于 2D 空间卷积是 CNN 在视觉任务中的基本操作，这里我们将其表示为普通卷积，并首先简要回顾它。2D 卷积中有两个主要步骤：1) 在输入特征图 x 上采样局部感受野区域 R；2) 通过加权求和聚合采样值。因此，输出特征图 y 可以表示为

$$y(p) = \sum_{p\in R} w(p) \cdot x(p+p), \quad (1)$$

其中 p 表示输入和输出特征图上的当前位置，而 p 枚举 R 中的位置。例如，使用 3×3 内核和膨胀 1 的卷积操作的局部感受野区域是 $R = \{(-1,-1), (-1,0), \cdots, (0,1), (1,1)\}$。

香草卷积遇见中心差分。在受著名的局部二值模式（LBP）[7]的启发，该模式以二值中心差分方式描述局部关系，我们也把中心差分引入到基础卷积中，以增强其表征能力和泛化能力。类似地，中心差分卷积也包含两个步骤，即采样和聚合。采样步骤与基础卷积中的步骤相似，而聚合步骤则不同：如图 2 所示，中心差分卷积倾向于聚合采样值的中心导向梯度。式(1)变为

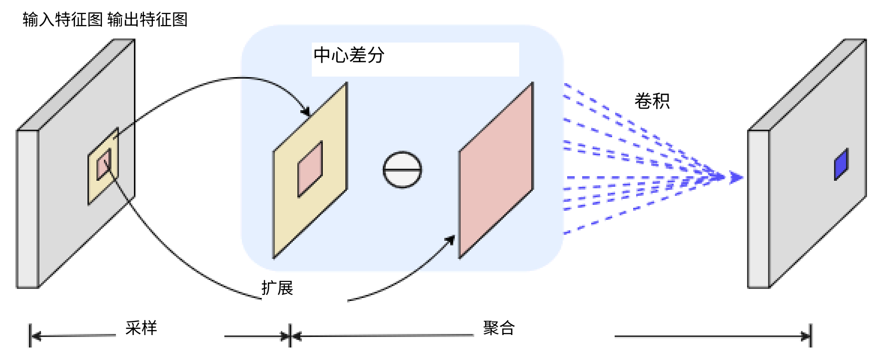

图 2. 中心差分卷积。

$$y(p) = \sum_{p\in R} w(p) \cdot (x(p+p) - x(p)). \quad (2)$$

当 p= (0, 0) 时，梯度值相对于中心位置本身始终等于零。在人脸防伪任务中，强度层语义信息和梯度层详细信息对于区分真实和伪造人脸都至关重要，这表明将普通卷积与中心差分卷积相结合可能是一种可行的方法，以提供更强的建模能力。因此我们将中心差分卷积推广为

$$\begin{aligned} y(p) = \theta \cdot & \underbrace{\sum_{p\in R} w(p) \cdot (x(p+p) - x(p))}_{\text{中心差分卷积}} \\ & + (1-\theta) \cdot \underbrace{\sum_{p\in R} w(p) \cdot x(p+p)}_{\text{香草卷积}}, \end{aligned} \quad (3)$$

其中超参数 θ ∈ [0, 1] 在强度级信息和梯度级信息之间进行权衡。θ 的值越高，表示中心差分梯度信息的重要性越大。我们在此后称这种广义中心差分卷积为 CDC，根据其上下文应易于识别。

CDC 的实现。为了在现代深度学习框架中高效实现 CDC，我们将公式(3)分解并合并为普通的卷积，并添加中心差分项

$$y(p) = \underbrace{\sum_{p\in R} w(p) \cdot x(p+p)}_{\text{vanilla convolution}} + \theta \cdot \underbrace{\left(-x(p) \cdot \sum_{p\in R} w(p)\right)}_{\text{中心差分项}}. \quad (4)$$

根据公式(4)，CDC 可以通过几行 PyTorch[42]和 TensorFlow[1]的代码轻松实现。公式(4)的推导和基于 PyTorch 的代码在附录 A 中展示。

与先前工作的关系。在这里，我们讨论 CDC 与普通卷积、局部二值卷积[27]和 Gabor 卷积[38]之间的关系，它们共享

similar design philosophy but with different focuses. The ablation study is in Section 4.3 to show superior performance of CDC for face anti-spoofing task.

Relation to Vanilla Convolution. CDC is more generalized. It can be seen from Eq. (3) that vanilla convolution is a special case of CDC when $\theta = 0$, i.e., aggregating local intensity information without gradient message.

Relation to Local Binary Convolution[27]. Local binary convolution (LBConv) focuses on computational reduction so its modeling capacity is limited. CDC focuses on enhancing rich detailed feature representation capacity without any additional parameters. On the other side, LBConv uses pre-defined filters to describe the local feature relation while CDC can learn these filters automatically.

Relation to Gabor Convolution[38]. Gabor convolution (GaborConv) devotes to enhancing the representation capacity of spatial transformations (i.e., orientation and scale changes) while CDC focuses more on representing fine-grained robust features in diverse environments.

3.2. CDCN

Depth-supervised face anti-spoofing methods [36, 4] take advantage of the discrimination between spoofing and living faces based on 3D shape, and provide pixel-wise detailed information for the FAS model to capture spoofing cues. Motivated by this, a similar depth-supervised network [36] called "DepthNet" is built up as baseline in this paper. In order to extract more fine-grained and robust features for estimating the facial depth map, CDC is introduced to form **C**entral **D**ifference **C**onvolutional **N**etworks (CDCN). Note that DepthNet is the special case of the proposed CDCN when $\theta = 0$ for all CDC operators.

The details of CDCN are shown in Table 1. Given a single RGB facial image with size $3 \times 256 \times 256$, multi-level (low-level, mid-level and high-level) fused features are extracted for predicting the grayscale facial depth with size 32×32. We use $\theta = 0.7$ as the default setting and ablation study about θ will be shown in Section 4.3.

For the loss function, mean square error loss $\mathcal{L}_{MSE}$ is utilized for pixel-wise supervision. Moreover, for the sake of fine-grained supervision needs in FAS task, contrastive depth loss $\mathcal{L}_{CDL}$ [56] is considered to help the networks learn more detailed features. So the overall loss $L_{overall}$ can be formulated as $\mathcal{L}_{overall} = \mathcal{L}_{MSE} + \mathcal{L}_{CDL}$.

3.3. CDCN++

It can be seen from Table 1 that the architecture of CDCN is designed coarsely (e.g., simply repeating the same block structure for different levels), which might be sub-optimized for face anti-spoofing task. Inspired by the classical visual object understanding models [40], we propose an extended version CDCN++ (see Fig. 5), which consists of a NAS based backbone and Multiscale Attention Fusion Module (MAFM) with selective attention capacity.

Level	Output	DepthNet [36]	CDCN ($\theta = 0.7$)
	256×256	3×3 conv, 64	3×3 **CDC**, 64
Low	128×128	3×3 conv, 128 3×3 conv, 196 3×3 conv, 128 3×3 max pool	3×3 **CDC**, 128 3×3 **CDC**, 196 3×3 **CDC**, 128 3×3 max pool
Mid	64×64	3×3 conv, 128 3×3 conv, 196 3×3 conv, 128 3×3 max pool	3×3 **CDC**, 128 3×3 **CDC**, 196 3×3 **CDC**, 128 3×3 max pool
High	32×32	3×3 conv, 128 3×3 conv, 196 3×3 conv, 128 3×3 max pool	3×3 **CDC**, 128 3×3 **CDC**, 196 3×3 **CDC**, 128 3×3 max pool
	32×32	[concat (Low, Mid, High), 384]	
	32×32	3×3 conv, 128 3×3 conv, 64 3×3 conv, 1	3×3 **CDC**, 128 3×3 **CDC**, 64 3×3 **CDC**, 1
# params		2.25×10^6	2.25×10^6

Table 1. Architecture of DepthNet and CDCN. Inside the brackets are the filter sizes and feature dimensionalities. "conv" and "CDC" suggest vanilla and central difference convolution, respectively. All convolutional layers are with stride=1 and followed by a BN-ReLU layer while max pool layers are with stride=2.

Search Backbone for FAS task. Our searching algorithm is based on two gradient-based NAS methods [35, 60], and more technical details can be referred to the original papers. Here we mainly state the new contributions about searching backbone for FAS task.

As illustrated in Fig. 3(a), the goal is to search for cells in three levels (low-level, mid-level and high-level) to form a network backbone for FAS task. Inspired by the dedicated neurons for hierarchical organization in human visual system [40], we prefer to search these multi-level cells freely (i.e., cells with varied structures), which is more flexible and generalized. We name this configuration as "**Varied Cells**" and will study its impacts in Sec. 4.3 (see Tab. 2). Different from previous works [35, 60], we adopt only one output of the latest incoming cell as the input of the current cell.

As for the cell-level structure, Fig. 3(b) shows that each cell is represented as a directed acyclic graph (DAG) of N nodes $\{x\}_{i=0}^{N-1}$, where each node represents a network layer. We denote the operation space as $\mathcal{O}$, and Fig. 3(c) shows eight designed candidate operations (none, skip-connect and CDCs). Each edge (i, j) of DAG represents the information flow from node x_i to node x_j, which consists of the candidate operations weighted by the architecture parameter $\alpha^{(i,j)}$. Specially, each edge (i, j) can be formulated by a function $\tilde{o}^{(i,j)}$ where $\tilde{o}^{(i,j)}(x_i) = \sum_{o \in \mathcal{O}} \eta_o^{(i,j)} \cdot o(x_i)$. Softmax function is utilized to relax architecture parameter $\alpha^{(i,j)}$ into operation weight $o \in \mathcal{O}$, that is $\eta_o^{(i,j)} = \frac{exp(\alpha_o^{(i,j)})}{\sum_{o' \in \mathcal{O}} exp(\alpha_{o'}^{(i,j)})}$. The intermediate node can be denoted as $x_j = \sum_{i<j} \tilde{o}^{(i,j)}(x_i)$ and the output node x_{N-1} is represented by weighted summation of all intermediate nodes $x_{N-1} = \sum_{0<i<N-1} \beta_i(x_i)$. Here we propose a node at-

采用相似的设计理念但侧重点不同。消融研究在 4.3 节中，以展示 CDC 在人脸反欺骗任务上的优越性能。

与普通卷积的关系。CDC 更为通用。
已实现。从公式(3)可以看出，普通卷积是 CDC 在θ=0 时的一个特例，即仅聚合局部强度信息而不传递梯度信息。

与局部二值卷积[27]的关系。局部二值卷积
卷积（LBConv）专注于计算减少，因此其建模能力有限。CDC 专注于增强丰富的细节特征表示能力，而不需要任何额外的参数。另一方面，LBConv 使用预定义的滤波器来描述局部特征关系，而 CDC 可以自动学习这些滤波器。

与 Gabor 卷积[38]的关系。Gabor 卷积(GaborConv)致力于提升空间变换（即方向和尺度变化）的表征能力，而 CDC 则更专注于在不同环境中表征细粒度的鲁棒特征。

3.2. CDCN 深度监督人脸防伪方法[36, 4]利用了伪造人脸和真实人脸在 3D 形状上的差异，为 FAS 模型提供像素级的详细信息以捕捉伪造线索。受此启发，本文构建了一个类似的深度监督网络[36]，称为“DepthNet”，作为基线。为了提取更细粒度和鲁棒的特征以估计面部深度图，引入了 CDC 形成中心差分卷积网络（CDCN）。请注意，当所有 CDC 算子θ = 0 时，DepthNet 是所提出的 CDCN 的特殊情况。

CDCN 的细节如表 1 所示。给定一个大小为 3 × 256 × 256 的单个 RGB 人脸图像，提取多级（低级、中级和高级）融合特征以预测大小为 32 × 32 的灰度人脸深度。我们使用θ = 0.7 作为默认设置，θ的消融研究将在第 4.3 节中展示。

对于损失函数，像素级监督采用均方误差损失 Lis。此外，为了满足 FAS 任务中细粒度监督的需求，考虑使用对比深度损失 L[56]来帮助网络学习更详细的特征。因此，整体损失 L 可以表示为 L= L+ L。

3.3. CDCN++ 从表 1 可以看出，CDCN 的架构设计较为粗略（例如，对不同层级简单地重复相同的块结构），这可能对面部反欺骗任务来说不是最优的。受经典的视觉对象理解模型[40]的启发，我们提出了一个扩展版本 CDCN++（见图 5），它由基于 NAS 的主干和 Multiscale Attention Fusion 组成。

Level	Output	DepthNet [36]	CDCN (θ = 0.7)
	256 × 256	3 × 3 conv, 64	3 × 3 CDC, 64
低	128 × 128	3 × 3 卷积，128 3 × 3 卷积，196 3 × 3 卷积，128 3 × 3 最大池化	3 × 3 CDC, 128 3 × 3 CDC, 196 3 × 3 CDC, 128 3 × 3 最大池化
	64 × 64	3 × 3 卷积，128 3 × 3 卷积，196 3 × 3 卷积，128 3 × 3 最大池化	3 × 3 CDC, 128 3 × 3 CDC, 196 3 × 3 CDC, 128 3 × 3 最大池化
高	32 × 32	3 × 3 卷积，128 3 × 3 卷积，196 3 × 3 卷积，128 3 × 3 最大池化	3 × 3 CDC, 128 3 × 3 CDC, 196 3 × 3 CDC, 128 3 × 3 最大池化
	32 × 32	[concat (低, 中, 高), 384]	
	32 × 32	3 × 3 卷积，128 3 × 3 卷积，64 3 × 3 卷积，1	3 × 3 CDC, 128 3 × 3 CDC, 64 3 × 3 CDC, 1
# params		2.25 × 10	2.25 × 10

表 1. DepthNet 和 CDCN 的架构。括号内是滤波器尺寸和特征维度。“conv”和“CDC”分别表示普通卷积和中心差分卷积。所有卷积层步长为 1，并后接 BN-ReLU 层，而最大池化层步长为 2。

具有选择性注意力能力的模块（MAFM）。

搜索用于 FAS 任务的骨干网络。我们的搜索算法
该算法基于两种基于梯度的 NAS 方法[35, 60]，更多技术细节可参考原始论文。这里我们主要陈述关于为 FAS 任务搜索骨干网络的新贡献。

如图 3(a)所示，目标是在三个层次（低级、中级和高级）中搜索细胞，以形成 FAS 任务的网络骨干。受人类视觉系统中用于层次组织的专用神经元[40]的启发，我们倾向于自由搜索这些多级细胞（即具有不同结构的细胞），这更加灵活和泛化。我们将这种配置命名为“多样化细胞”，并将研究其在第 4.3 节的影响（参见表 2）。与之前的工作[35, 60]不同，我们仅将最新输入细胞的单个输出作为当前细胞的输入。

关于单元级结构，图 3(b)显示每个单元被表示为包含 N 个节点{x}i 的有向无环图（DAG），其中每个节点代表一个网络层。我们将操作空间记为 O，图 3(c)展示了设计的八个候选操作（无操作、跳接连接和 CDCs）。DAG 的每条边(i, j)表示从节点 x 流向节点 x 的信息流，该信息流由候选操作加权，权重由架构参数α决定。特别地，每条边(i, j)可以表示为一个函数˜o，其中˜o(x) =

$$\Sigma_{o\in O}\, \eta_o^{(i,j)} \cdot o(x)。$$

Softmax 函数用于将架构参数α松弛为操作权重 o ∈ O，即ηo =

$\frac{\exp(\alpha_o)}{\Sigma_{o'\in O}\exp(\alpha_o)}$. 中间节点可以表示为

ẋ= Σ

Σ 0

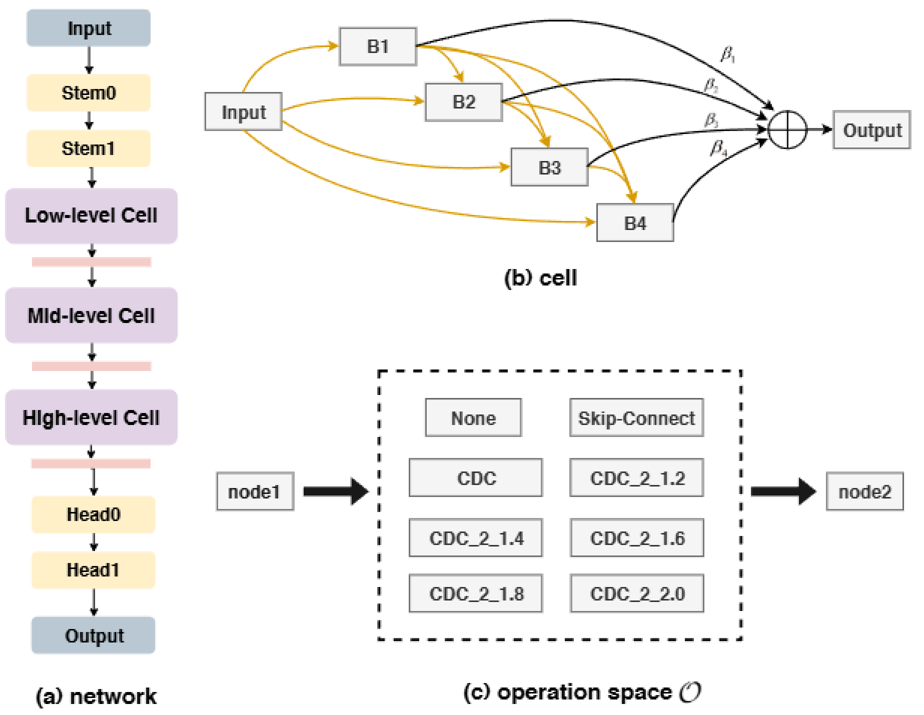

Figure 3. Architecture search space with CDC. (a) A network consists of three stacked cells with max pool layer while stem and head layers adopt CDC with 3×3 kernel and $\theta = 0.7$. (b) A cell contains 6 nodes, including an input node, four intermediate nodes B1, B2, B3, B4 and an output node. (c) The edge between two nodes (except output node) denotes a possible operation. The operation space consists of eight candidates, where **CDC_2_**r means using two stacked CDC to increase channel number with ratio r first and then decrease back to the original channel size. The size of total search space is $3 \times 8^{(1+2+3+4)} = 3 \times 8^{10}$.

tention strategy to learn the importance weights β among intermediate nodes, that is $\beta_i = \frac{exp(\beta_i')}{\sum_{0<j<N-1} exp(\beta_j')}$, where β_i is the softmax of the original learnable weight $\beta_i{}'$ for intermediate node x_i.

In the searching stage, $\mathcal{L}_{train}$ and $\mathcal{L}_{val}$ are denoted as the training and validation loss respectively, which are all based on the depth-supervised loss $\mathcal{L}_{overall}$ described in Sec. 3.2. Network parameters w and architecture parameters α are learned with the following bi-level optimization problem:

$$\begin{aligned} &\min_{\alpha} \quad \mathcal{L}_{val}(w^*(\alpha), \alpha), \\ &s.t. \quad w^*(\alpha) = arg\min_{w} \mathcal{L}_{train}(w, \alpha) \end{aligned} \tag{5}$$

After convergence, the final discrete architecture is derived by: 1) setting $o^{(i,j)} = arg\,max_{o \in \mathcal{O}, o \neq none}\, p_o^{(i,j)}$; 2) for each intermediate node, choosing one incoming edge with the largest value of $max_{o \in \mathcal{O}, o \neq none}\, p_o^{(i,j)}$, and 3) for each output node, choosing the one incoming intermediate node with largest value of $max_{0<i<N-1}\, \beta_i$ (denoted as "**Node Attention**") as input. In contrast, choosing last intermediate node as output node is more straightforward. We will compare these two settings in Sec. 4.3 (see Tab. 2).

MAFM. Although simply fusing low-mid-high levels features can boost performance for the searched CDC architecture, it is still hard to find the important regions to focus, which goes against learning more discriminative features. Inspired by the selective attention in human visual system

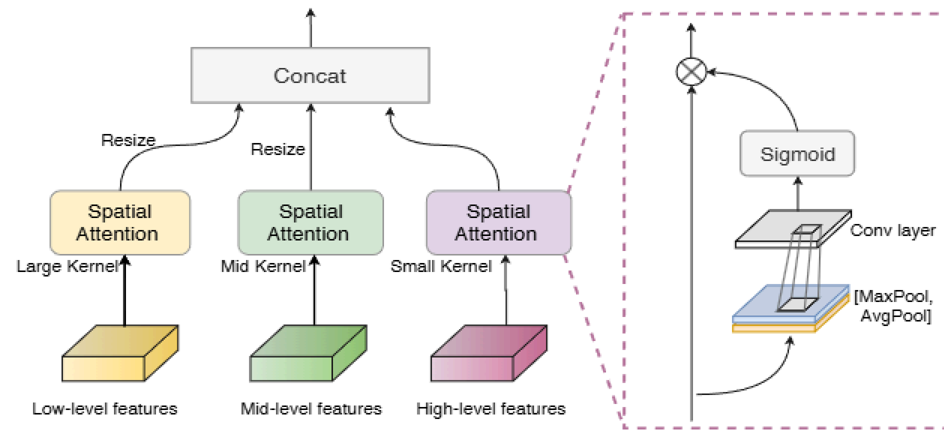

Figure 4. Multiscale Attention Fusion Module.

[40, 55], neurons at different levels are likely to have stimuli in their receptive fields with various attention. Here we propose a Multiscale Attention Fusion Module (MAFM), which is able to refine and fuse low-mid-high levels CDC features via spatial attention.

As illustrated in Fig. 4, features $\mathcal{F}$ from different levels are refined via spatial attention [58] with receptive fields related kernel size (i.e., the high/semantic level should be with small attention kernel size while low level with large attention kernel size in our case) and then concatenate together. The refined features $\mathcal{F}'$ can be formulated as

$$\mathcal{F}_i' = \mathcal{F}_i \odot (\sigma(\mathcal{C}_i([\mathcal{A}(\mathcal{F}_i), \mathcal{M}(\mathcal{F}_i)]))), i \in \{low, mid, high\}, \tag{6}$$

where $\odot$ represents Hadamard product. $\mathcal{A}$ and $\mathcal{M}$ denotes avg and max pool layer respectively. σ means the sigmoid function while $\mathcal{C}$ is the convolution layer. Vanilla convolutions with 7×7, 5×5 and 3×3 kernels are utilized for $\mathcal{C}_{low}$, $\mathcal{C}_{mid}$ and $\mathcal{C}_{high}$, respectively. CDC is not chosen here because of its limited capacity of global semantic cognition, which is vital in spatial attention. The corresponding ablation study is conducted in Sec. 4.3.

4. Experiments

In this section, extensive experiments are performed to demonstrate the effectiveness of our method. In the following, we sequentially describe the employed datasets & metrics (Sec. 4.1), implementation details (Sec. 4.2), results (Sec. 4.3 - 4.5) and analysis (Sec. 4.6).

4.1. Datasets and Metrics

Databases. Six databases OULU-NPU [10], SiW [36], CASIA-MFSD [66], Replay-Attack [13], MSU-MFSD [57] and SiW-M [37] are used in our experiments. OULU-NPU and SiW are high-resolution databases, containing four and three protocols to validate the generalization (e.g., unseen illumination and attack medium) of models respectively, which are utilized for intra testing. CASIA-MFSD, Replay-Attack and MSU-MFSD are databases which contain low-resolution videos, which are used for cross testing. SiW-M is designed for cross-type testing for unseen attacks as there are rich (13) attacks types inside.

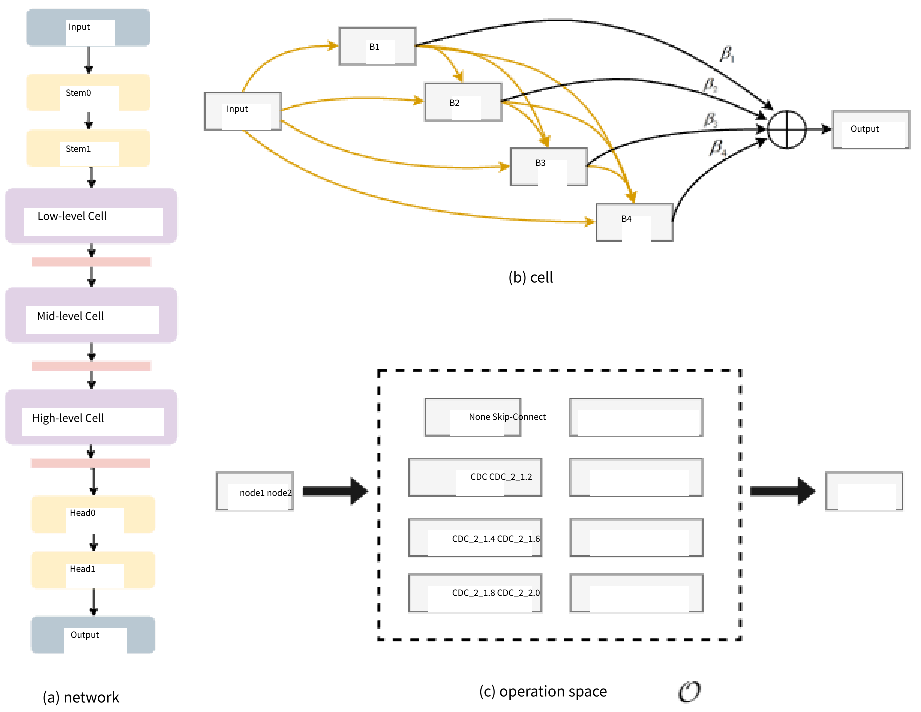

图 3. 带有 CDC 的架构搜索空间。（a）一个网络由三个堆叠的单元组成，其中包含最大池化层，而干层和头层采用 3×3 内核和θ=0.7 的 CDC。（b）一个单元包含 6 个节点，包括一个输入节点、四个中间节点 B1、B2、B3、B4 和一个输出节点。（c）两个节点（除输出节点外）之间的边表示一个可能的操作。操作空间包含八个候选，其中 CDC 2 r 表示使用两个堆叠的 CDC 来增加通道数，先以比例 r 增加，然后减少回原始通道大小。总搜索空间的大小为 3 × 8= 3 × 8。

注意力策略用于学习中间节点的重要性权重β，即β=Σ，其中β是中间节点 x 的原可学习权重β的 softmax。

在搜索阶段，Land L 分别表示训练损失和验证损失，它们都基于第 3.2 节中描述的深度监督损失 L。网络参数 w 和架构参数α通过以下双层优化问题进行学习：

$$\min_{\alpha} \quad L(w(\alpha), \alpha), \qquad \text{s.t. } w(\alpha) = \arg\min_{w} \quad L(w, \alpha) \tag{5}$$

收敛后，最终离散架构由以下步骤导出：1) 设置 o= arg maxpo；2) 对每个中间节点，选择具有最大 maxpo 值的入边；3) 对每个输出节点，选择具有最大 maxβ 的入边中间节点（记为“节点注意力”）作为输入。相比之下，将最后一个中间节点作为输出节点更为直接。我们将在第 4.3 节（参见表 2）比较这两种设置。

MAFM。尽管简单地融合低-中-高层数据特征可以提升所搜索的 CDC 架构的性能，但仍难以找到需要关注的重要区域，这与学习更具判别性的特征背道而驰。受人类视觉系统中选择性注意力的启发

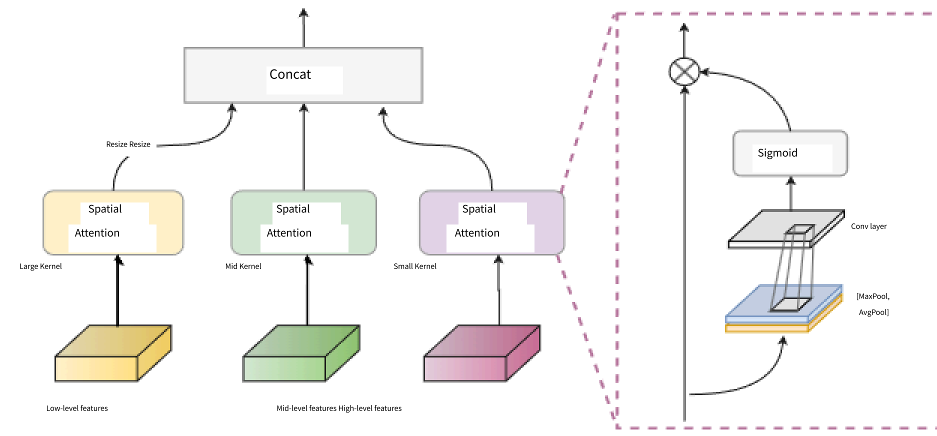

图 4. 多尺度注意力融合模块。

[40, 55]，不同层级的神经元在其感受野中可能具有不同注意力的刺激。这里我们提出一个多尺度注意力融合模块（MAFM），它能够通过空间注意力来精炼和融合低-中-高层的 CDC 特征。

如图 4 所示，来自不同层级的特征 F 通过空间注意力 [58] 进行精炼，其感受野与相关的内核大小相关（即，在我们的案例中，高层/语义层应使用较小的注意力内核大小，而低层则使用较大的注意力内核大小），然后连接在一起。精炼后的特征 F 可以表示为

$$F= F(\sigma(C([A(F), M(F)]))), i \in \{low, mid, high\} \tag{6}$$

where 表示 Hadamard 乘积。A 和 M 分别表示 avg 池化层和 max 池化层。σ表示 sigmoid 函数，而 C 是卷积层。C、C 和 C 分别使用 7×7、5×5 和 3×3 核的普通卷积。这里没有选择 CDC，因为其在全局语义认知方面的能力有限，而全局语义认知在空间注意力中至关重要。相应的消融实验在 4.3 节中进行。

4. 实验 在本节中，我们进行了大量实验以验证我们方法的有效性。在以下内容中，我们依次描述所使用的数据集和指标（4.1 节）、实现细节（4.2 节）、结果（4.3-4.5 节）和分析（4.6 节）。

4.1. 数据集和指标数据库。我们的实验使用了六个数据库：OULU-NPU [10]、SiW [36]、CASIA-MFSD [66]、Replay-Attack [13]、MSU-MFSD [57] 和 SiW-M [37]。OULU-NPU 和 SiW 是高分辨率数据库，分别包含四个和三个协议，用于验证模型的泛化能力（例如，未知的照明条件和攻击媒介），这些数据库用于内部测试。CASIA-MFSD、ReplayAttack 和 MSU-MFSD 是包含低分辨率视频的数据库，用于交叉测试。SiW-M 是为交叉类型测试设计的，用于未知攻击，因为其中包含丰富的（13）攻击类型。

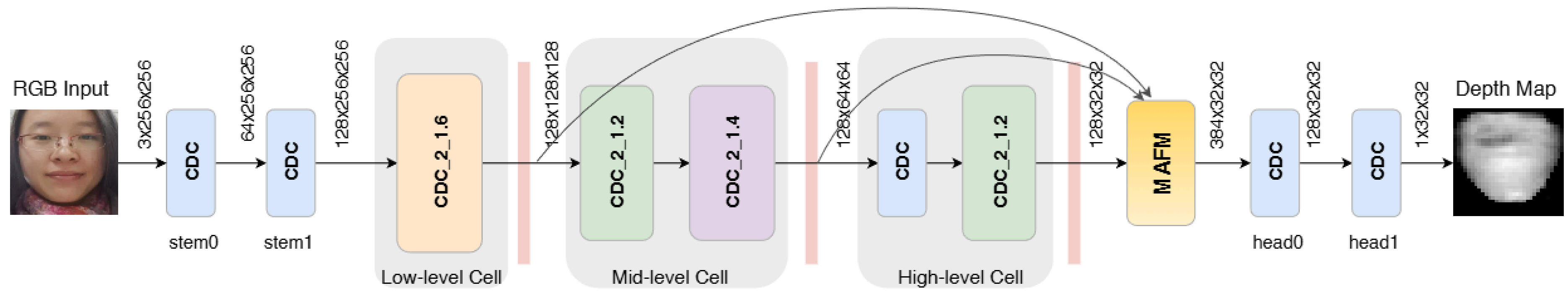

Figure 5. The architecture of CDCN++. It consists of the searched CDC backbone and MAFM. Each cell is followed by a max pool layer.

Performance Metrics. In OULU-NPU and SiW dataset, we follow the original protocols and metrics, i.e., Attack Presentation Classification Error Rate (APCER), Bona Fide Presentation Classification Error Rate (BPCER), and ACER [24] for a fair comparison. Half Total Error Rate (HTER) is adopted in the cross testing between CASIA-MFSD and Replay-Attack. Area Under Curve (AUC) is utilized for intra-database cross-type test on CASIA-MFSD, Replay-Attack and MSU-MFSD. For the cross-type test on SiW-M, APCER, BPCER, ACER and Equal Error Rate (EER) are employed.

4.2. Implementation Details

Depth Generation. Dense face alignment PRNet [18] is adopted to estimate the 3D shape of the living face and generate the facial depth map with size 32×32. More details and samples can be found in [56]. To distinguish living faces from spoofing faces, at the training stage, we normalize living depth map in a range of $[0, 1]$, while setting spoofing depth map to 0, which is similar to [36].

Training and Testing Setting. Our proposed method is implemented with Pytorch. In the training stage, models are trained with Adam optimizer and the initial learning rate (lr) and weight decay (wd) are 1e-4 and 5e-5, respectively. We train models with maximum 1300 epochs while lr halves every 500 epochs. The batch size is 56 on eight 1080Ti GPUs. In the testing stage, we calculate the mean value of the predicted depth map as the final score.

Searching Setting. Similar to [60], partial channel connection and edge normalization are adopted. The initial number of channel is sequentially $\{32, 64, 128, 128, 128, 64, 1\}$ in the network (see Fig. 3(a)), which doubles after searching. Adam optimizer with lr=1e-4 and wd=5e-5 is utilized when training the model weights. The architecture parameters are trained with Adam optimizer with lr=6e-4 and wd=1e-3. We search 60 epochs on Protocol-1 of OULU-NPU with batchsize 12 while architecture parameters are not updated in the first 10 epochs. The whole process costs one day on three 1080Ti.

4.3. Ablation Study

In this subsection, all ablation studies are conducted on Protocol-1 (different illumination condition and location between train and test sets) of OULU-NPU [10] to explore the details of our proposed CDC, CDCN and CDCN++.

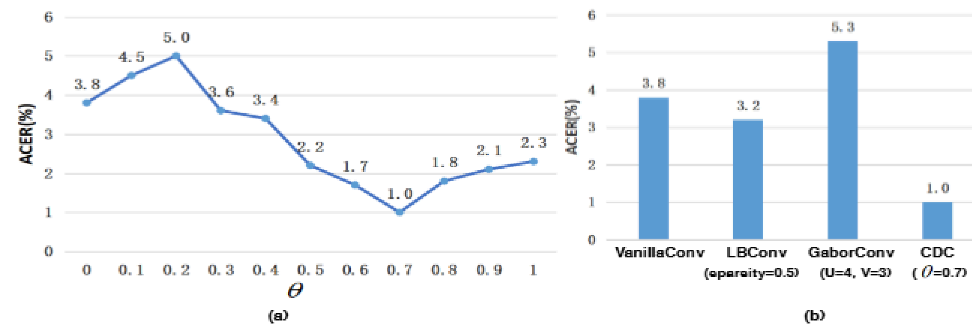

Figure 6. (a) Impact of θ in CDCN. (b) Comparison among various convolutions (only showing the hyperparameters with best performance). The lower ACER, the better performance.

Table 2. The ablation study of NAS configuration.

Model	Varied cells	Node attention	ACER(%)
NAS_Model 1			1.7
NAS_Model 2	√		1.5
NAS_Model 3		√	1.4
NAS_Model 4	√	√	**1.3**

Impact of θ in CDCN. According to Eq. (3), θ controls the contribution of the gradient-based details, i.e., the higher θ, the more local detailed information included. As illustrated in Fig. 6(a), when $\theta \geqslant 0.3$, CDC always achieves better performance than vanilla convolution ($\theta = 0$, ACER=3.8%), indicating the central difference based fine-grained information is helpful for FAS task. As the best performance (ACER=1.0%) is obtained when $\theta = 0.7$, we use this setting for the following experiments. Besides keeping the constant θ for all layers, we also explore an adaptive CDC method to learn θ for every layer, which is shown in ***Appendix B***.

CDC vs. Other Convolutions. As discussed in Sec. 3.1 about the relation between CDC and prior convolutions, we argue that the proposed CDC is more suitable for FAS task as the detailed spoofing artifacts in diverse environments should be represented by the gradient-based invariant features. Fig. 6(b) shows that CDC outperforms other convolutions by a large margin (more than 2% ACER). It is interesting to find that LBConv performs better than vanilla convolution, indicating that the local gradient information is important for FAS task. GaborConv performs the worst because it is designed for capturing spatial invariant features, which is not helpful in face anti-spoofing task.

Impact of NAS Configuration. Table 2 shows the ablation study about the two NAS configurations described in Sec. 3.3, i.e., varied cells and node attention. Compared to the baseline setting with the shared cells and last inter-

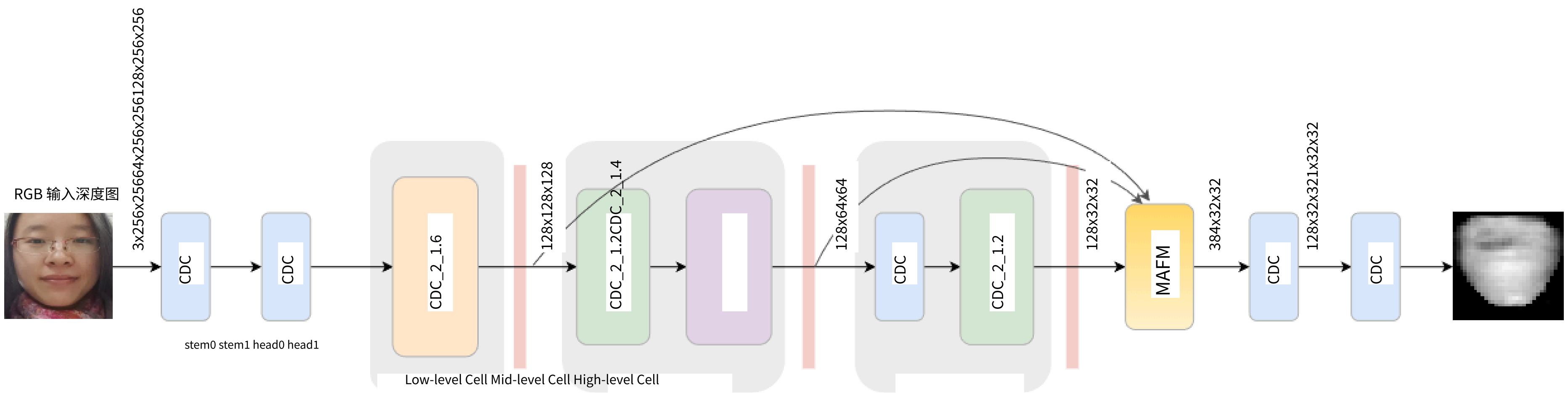

图 5. CDCN++的架构。它由搜索的 CDC 主干和 MAFM 组成。每个单元后跟一个最大池化层。

性能指标。在 OULU-NPU 和 SiW 数据集上，我们遵循原始协议和指标，即攻击呈现分类错误率（APCER）、真实呈现分类错误率（BPCER）和 ACER [24]，以进行公平比较。在 CASIAMFSD 和 Replay-Attack 之间的交叉测试中采用半总错误率（HTER）。在 CASIA-MFSD、Replay-Attack 和 MSU-MFSD 上的数据库内交叉类型测试中采用曲线下面积（AUC）。在 SiW-M 上的交叉类型测试中采用 APCER、BPCER、ACER 和等错误率（EER）。

4.2. 实现细节深度生成。采用稠密人脸对齐 PRNet [18]来估计真实人脸的 3D 形状，并生成 32 × 32 尺寸的人脸深度图。更多细节和样本可以在[56]中找到。为了区分真实人脸和伪造人脸，在训练阶段，我们将真实深度图归一化到[0, 1]范围内，而将伪造深度图设置为 0，这与[36]类似。

训练和测试设置。我们提出的方法使用 Pytorch 实现。在训练阶段，模型采用 Adam 优化器，初始学习率（lr）和权重衰减（wd）分别为 1e-4 和 5e-5。我们用最大 1300 个 epoch 训练模型，每 500 个 epoch 学习率减半。在八个 1080Ti GPU 上，批处理大小为 56。在测试阶段，我们将预测深度图的平均值作为最终得分。

搜索设置。类似于[60]，采用部分通道连接和边缘归一化。网络中的初始通道数按顺序为{32, 64, 128, 128, 128, 64, 1}（见图 3(a)），搜索后通道数翻倍。使用 Adam 优化器，学习率 lr=1e-4，权重衰减 wd=5e-5 来训练模型权重。架构参数使用 Adam 优化器，学习率 lr=6e-4，权重衰减 wd=1e-3 进行训练。我们在 OULU-NPU 的 Protocol-1 上搜索了 60 个 epoch，批量大小为 12，而前 10 个 epoch 中架构参数不进行更新。整个过程在三个 1080Ti 上花费了一天时间。

4.3. 消融研究 在本小节中，所有消融研究均在 OULU-NPU [10]的协议-1（训练集和测试集之间的光照条件和位置不同）上进行，以探索

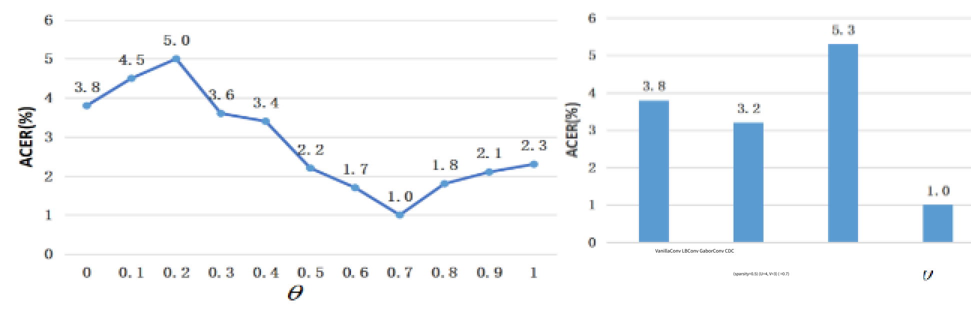

图 6. (a) CDCN 中θ的影响。(b) 各种卷积的比较（仅显示性能最佳的超参数）。ACER 越低，性能越好。

表 2. NAS 配置的消融研究。

模型	变异细胞	节点注意力	ACER(%)
NAS 模型 1			1.7
NAS 模型 2	√		1.5
NAS 模型 3		√	1.4
NAS 模型 4	√	√	1.3

我们提出的 CDC、CDCN 和 CDCN++的详细情况。

CDCN 中θ的影响。根据公式(3)，θ控制基于梯度的细节的贡献，即θ越高，包含的局部细节信息越多。如图 6(a)所示，当θ > 0.3 时，CDC 始终比普通卷积（θ = 0，ACER=3.8%）表现更好，这表明基于中心差分的细粒度信息对 FAS 任务有帮助。当θ = 0.7 时获得最佳性能（ACER=1.0%），因此我们在后续实验中使用此设置。除了对所有层保持恒定的θ，我们还探索了一种自适应 CDC 方法来为每一层学习θ，这显示在附录 B 中。

CDC 与其他卷积。如第几节所述。3.1 关于 CDC 与先前卷积的关系，我们认为所提出的 CDC 更适合 FAS 任务，因为不同环境中的详细欺骗伪影应该由基于梯度的不变特征来表示。图 6(b)显示，CDC 比其他卷积方法大幅领先（ACER 超过 2%）。有趣的是发现，LBConv 的表现优于普通卷积，这表明局部梯度信息对 FAS 任务很重要。GaborConv 表现最差，因为它设计用于捕获空间不变特征，这在面部反欺骗任务中并不有帮助。

NAS 配置的影响。表 2 显示了一项关于第 3.3 节中描述的两个 NAS 配置的研究，即变化单元和节点注意力。与共享单元和最后交互的基线设置相比

Table 3. The ablation study of NAS based backbone and MAFM.

Backbone	Multi-level Fusion	ACER(%)
w/o NAS	w/ multi-level concat	1.0
w/ NAS	w/ multi-level concat	0.7
w/ NAS	w/ MAFM (3x3,3x3,3x3 CDC)	1.2
w/ NAS	w/ MAFM (3x3,3x3,3x3 VanillaConv)	0.6
w/ NAS	w/ MAFM (5x5,5x5,5x5 VanillaConv)	1.1
w/ NAS	w/ MAFM (7x7,5x5,3x3 VanillaConv)	**0.2**

Table 4. The results of intra testing on four protocols of OULU-NPU. We only report the results "STASN [62]" trained without extra datasets for a fair comparison.

Prot.	Method	APCER(%)	BPCER(%)	ACER(%)
1	GRADIANT [6]	1.3	12.5	6.9
	STASN [62]	1.2	2.5	1.9
	Auxiliary [36]	1.6	1.6	1.6
	FaceDs [26]	1.2	1.7	1.5
	FAS-TD [56]	2.5	0.0	1.3
	DeepPixBiS [20]	0.8	0.0	0.4
	CDCN (Ours)	0.4	1.7	1.0
	CDCN++ (Ours)	0.4	0.0	**0.2**
2	DeepPixBiS [20]	11.4	0.6	6.0
	FaceDs [26]	4.2	4.4	4.3
	Auxiliary [36]	2.7	2.7	2.7
	GRADIANT [6]	3.1	1.9	2.5
	STASN [62]	4.2	0.3	2.2
	FAS-TD [56]	1.7	2.0	1.9
	CDCN (Ours)	1.5	1.4	1.5
	CDCN++ (Ours)	1.8	0.8	**1.3**
3	DeepPixBiS [20]	11.7±19.6	10.6±14.1	11.1±9.4
	FAS-TD [56]	5.9±1.9	5.9±3.0	5.9±1.0
	GRADIANT [6]	2.6±3.9	5.0±5.3	3.8±2.4
	FaceDs [26]	4.0±1.8	3.8±1.2	3.6±1.6
	Auxiliary [36]	2.7±1.3	3.1±1.7	2.9±1.5
	STASN [62]	4.7±3.9	0.9±1.2	2.8±1.6
	CDCN (Ours)	2.4±1.3	2.2±2.0	2.3±1.4
	CDCN++ (Ours)	1.7±1.5	2.0±1.2	**1.8±0.7**
4	DeepPixBiS [20]	36.7±29.7	13.3±14.1	25.0±12.7
	GRADIANT [6]	5.0±4.5	15.0±7.1	10.0±5.0
	Auxiliary [36]	9.3±5.6	10.4±6.0	9.5±6.0
	FAS-TD [56]	14.2±8.7	4.2±3.8	9.2±3.4
	STASN [62]	6.7±10.6	8.3±8.4	7.5±4.7
	FaceDs [26]	1.2±6.3	6.1±5.1	5.6±5.7
	CDCN (Ours)	4.6±4.6	9.2±8.0	6.9±2.9
	CDCN++ (Ours)	4.2±3.4	5.8±4.9	**5.0±2.9**

mediate node as output node, both these two configurations can boost the searching performance. The reason behind is twofold: 1) with more flexible searching constraints, NAS is able to find dedicated cells for different levels, which is more similar to human visual system [40], and 2) taking the last intermediate node as output might not be optimal while choosing the most important one is more reasonable.

Effectiveness of NAS Based Backbone and MAFM. The proposed CDCN++, consisting of NAS based backbone and MAFM, is shown in Fig. 5. It is obvious that cells from multiple levels are quite different and the mid-level cell has deeper (four CDC) layers. Table 3 shows the ablation studies about NAS based backbone and MAFM. It can be seen from the first two rows that NAS based backbone with direct multi-level fusion outperforms (0.3% ACER) the backbone without NAS, indicating the effectiveness of our searched architecture. Meanwhile, backbone with MAFM achieves 0.5% ACER lower than that with direct multi-level fusion, which shows the effectiveness of MAFM. We also analyse the convolution type and kernel size in MAFM and find that vanilla convolution is more suitable for capturing the semantic spatial attention. Besides, the attention kernel size should be large (7x7) and small (3x3) enough for low-level and high-level features, respectively.

Table 5. The results of intra testing on three protocols of SiW [36].

Prot.	Method	APCER(%)	BPCER(%)	ACER(%)
1	Auxiliary [36]	3.58	3.58	3.58
	STASN [62]	–	–	1.00
	FAS-TD [56]	0.96	0.50	0.73
	CDCN (Ours)	0.07	0.17	**0.12**
	CDCN++ (Ours)	0.07	0.17	**0.12**
2	Auxiliary [36]	0.57±0.69	0.57±0.69	0.57±0.69
	STASN [62]	–	–	0.28±0.05
	FAS-TD [56]	0.08±0.14	0.21±0.14	0.15±0.14
	CDCN (Ours)	0.00±0.00	0.13±0.09	0.06±0.04
	CDCN++ (Ours)	0.00±0.00	0.09±0.10	**0.04±0.05**
3	STASN [62]	–	–	12.10±1.50
	Auxiliary [36]	8.31±3.81	8.31±3.80	8.31±3.81
	FAS-TD [56]	3.10±0.81	3.09±0.81	3.10±0.81
	CDCN (Ours)	1.67±0.11	1.76±0.12	**1.71±0.11**
	CDCN++ (Ours)	1.97±0.33	1.77±0.10	1.90±0.15

4.4. Intra Testing

The intra testing is carried out on both the OULU-NPU and the SiW datasets. We strictly follow the four protocols on OULU-NPU and three protocols on SiW for the evaluation. All compared methods including STASN [62] are trained without extra datasets for a fair comparison.

Results on OULU-NPU. As shown in Table 4, our proposed CDCN++ ranks first on all 4 protocols (0.2%, 1.3%, 1.8% and 5.0% ACER, respectively), which indicates the proposed method performs well at the generalization of the external environment, attack mediums and input camera variation. Unlike other state-of-the-art methods (Auxiliary [36], STASN [62], GRADIANT [6] and FAS-TD [56]) extracting multi-frame dynamic features, our method needs only frame-level inputs, which is suitable for real-world deployment. It's worth noting that the NAS based backbone for CDCN++ is transferable and generalizes well on all protocols although it is searched on Protocol-1.

Results on SiW. Table 5 compares the performance of our method with three state-of-the-art methods Auxiliary [36], STASN [62] and FAS-TD [56] on SiW dataset. It can be seen from Table 5 that our method performs the best for all three protocols, revealing the excellent generalization capacity of CDC for (1) variations of face pose and expression, (2) variations of different spoof mediums, (3) cross/unknown presentation attack.

4.5. Inter Testing

To further testify the generalization ability of our model, we conduct cross-type and cross-dataset testing to verify the generalization capacity to unknown presentation attacks and unseen environment, respectively.

表 3. 基于 NAS 的主干网络和 MAFM 的消融研究

骨干	多层融合	ACER(%)
无 NAS	多层拼接	1.0
NAS	多层拼接	0.7
NAS	MAFM (3x3,3x3,3x3 CDC)	1.2
NAS	MAFM (3x3,3x3,3x3 VanillaConv)	0.6
NAS	MAFM (5x5,5x5,5x5 VanillaConv)	1.1
NAS	MAFM (7x7,5x5,3x3 VanillaConv)	0.2

表 4. 在 OULUNPU 的四个协议上进行的内部测试结果。我们仅报告“STASN [62]”的训练结果，以进行公平比较。

协议	方法	APCER(%)	BPCER(%)	ACER(%)
1	GRADIANT [6]	1.3	12.5	6.9
	STASN [62]	1.2	2.5	1.9
	Auxiliary [36]	1.6	1.6	1.6
	FaceDs [26]	1.2	1.7	1.5
	FAS-TD [56]	2.5	0.0	1.3
	DeepPixBiS [20]	0.8	0.0	0.4
	CDCN (我们)	0.4	1.7	1.0
	CDCN++ (我们)	0.4	0.0	0.2
2	DeepPixBiS [20]	11.4	0.6	6.0
	FaceDs [26]	4.2	4.4	4.3
	Auxiliary [36]	2.7	2.7	2.7
	GRADIANT [6]	3.1	1.9	2.5
	STASN [62]	4.2	0.3	2.2
	FAS-TD [56]	1.7	2.0	1.9
	CDCN (Ours)	1.5	1.4	1.5
	CDCN++ (Ours)	1.8	0.8	1.3
3	DeepPixBiS [20]	11.7±19.6	10.6±14.1	11.1±9.4
	FAS-TD [56]	5.9±1.9	5.9±3.0	5.9±1.0
	GRADIANT [6]	2.6±3.9	5.0±5.3	3.8±2.4
	FaceDs [26]	4.0±1.8	3.8±1.2	3.6±1.6
	Auxiliary [36]	2.7±1.3	3.1±1.7	2.9±1.5
	STASN [62]	4.7±3.9	0.9±1.2	2.8±1.6
	CDCN (Ours)	2.4±1.3	2.2±2.0	2.3±1.4
	CDCN++ (Ours)	1.7±1.5	2.0±1.2	1.8±0.7
4	DeepPixBiS [20]	36.7±29.7	13.3±14.1	25.0±12.7
	GRADIANT [6]	5.0±4.5	15.0±7.1	10.0±5.0
	Auxiliary [36]	9.3±5.6	10.4±6.0	9.5±6.0
	FAS-TD [56]	14.2±8.7	4.2±3.8	9.2±3.4
	STASN [62]	6.7±10.6	8.3±8.4	7.5±4.7
	FaceDs [26]	1.2±6.3	6.1±5.1	5.6±5.7
	CDCN (Ours)	4.6±4.6	9.2±8.0	6.9±2.9
	CDCN++ (Ours)	4.2±3.4	5.8±4.9	5.0±2.9

作为输出节点的中介节点，这两种配置都能提升搜索性能。其背后的原因有两个：1) 通过更灵活的搜索约束，NAS 能够为不同层级找到专用单元，这更类似于人类视觉系统[40]，2) 将最后一个中介节点作为输出可能并非最优，而选择最重要的一个则更为合理。

基于 NAS 的骨干网络和 MAFM 的有效性。

所提出的 CDCN++，由基于 NAS 的骨干网络和 MAFM 组成，如图 5 所示。很明显，来自多个层级的单元差异很大，中级单元具有更深（四个 CDC）层。表 3 展示了关于基于 NAS 的骨干网络和 MAFM 的消融研究。从前两行可以看出，具有直接多层级融合的基于 NAS 的骨干网络（ACER 提升 0.3%）优于没有 NAS 的骨干网络，这表明了我们搜索的架构的有效性。同时

具有 MAFM 的骨干网络实现了表 5。在 SiW[36]的三个协议上的内部测试结果。

	方法	APCER(%)	BPCER(%)	ACER(%)
1	辅助 [36]	3.58	3.58	3.58
	STASN [62]	–	–	1.00
	FAS-TD [56]	0.96	0.50	0.73
	CDCN (我们)	0.07	0.17	0.12
	CDCN++ (我们)	0.07	0.17	0.12
2	辅助 [36]	0.57±0.69	0.57±0.69	0.57±0.69
	STASN [62]	–	–	0.28±0.05
	FAS-TD [56]	0.08±0.14	0.21±0.14	0.15±0.14
	CDCN (我们)	0.00±0.00	0.13±0.09	0.06±0.04
	CDCN++ (我们)	0.00±0.00	0.09±0.10	0.04±0.05
3	STASN [62]	–	–	12.10±1.50
	辅助 [36]	8.31±3.81	8.31±3.80	8.31±3.81
	FAS-TD [56]	3.10±0.81	3.09±0.81	3.10±0.81
	CDCN (我们)	1.67±0.11	1.76±0.12	1.71±0.11
	CDCN++ (我们)	1.97±0.33	1.77±0.10	1.90±0.15

与直接多级融合相比，ACER 降低了 0.5%，这显示了 MAFM 的有效性。我们还分析了 MAFM 中的卷积类型和核大小，发现普通卷积更适合捕获语义空间注意力。此外，注意力核大小应足够大（7x7）和小（3x3），分别用于低级和高级特征。

4.4. 内部测试 内部测试在 OULU-NPU 和 SiW 数据集上进行。我们严格遵循 OULU-NPU 上的四个协议和 SiW 上的三个协议进行评估。所有比较方法包括 STASN [62] 都没有使用额外的数据集进行训练，以进行公平比较。

在 OULU-NPU 上的结果。如表 4 所示，我们提出的 CDCN++在所有 4 个协议（分别为 0.2%、1.3%、1.8%和 5.0%的 ACER）上均排名第一，这表明所提出的方法在外部环境、攻击媒介和输入摄像头变化方面具有良好的泛化能力。与其他最先进的方法（辅助[36]、STASN[62]、GRADIANT[6]和 FAS-TD[56]）提取多帧动态特征不同，我们的方法只需要帧级输入，适合实际部署。值得注意的是，CDCN++的 NAS 基础架构是可迁移的，并且在所有协议上都具有良好的泛化能力，尽管它是在协议-1 上搜索的。

在 SiW 上的结果。表 5 比较了我们的方法与三种最先进方法（辅助[36]、STASN[62]和 FAS-TD[56]）在 SiW 数据集上的性能。如表 5 所示，我们的方法在所有三个协议上表现最佳，揭示了 CDC 在（1）面部姿态和表情变化，（2）不同欺骗媒介变化，（3）跨/未知呈现攻击方面的出色泛化能力。

4.5. 交叉测试 为了进一步验证我们模型的泛化能力，我们进行了跨类型和跨数据集的测试，分别验证模型对未知呈现攻击和未见过环境的泛化能力。

Table 6. AUC (%) of the model cross-type testing on CASIA-MFSD, Replay-Attack, and MSU-MFSD.

Method	CASIA-MFSD [66]			Replay-Attack [13]			MSU-MFSD [57]			Overall
	Video	Cut Photo	Wrapped Photo	Video	Digital Photo	Printed Photo	Printed Photo	HR Video	Mobile Video	
OC-SVM$_{RBF}$+BSIF [3]	70.74	60.73	95.90	84.03	88.14	73.66	64.81	87.44	74.69	78.68±11.74
SVM$_{RBF}$+LBP [10]	91.94	91.70	84.47	99.08	98.17	87.28	47.68	99.50	97.61	88.55±16.25
NN+LBP [59]	94.16	88.39	79.85	99.75	95.17	78.86	50.57	99.93	93.54	86.69±16.25
DTN [37]	90.0	97.3	97.5	99.9	99.9	99.6	**81.6**	99.9	97.5	95.9±6.2
CDCN (Ours)	**98.48**	**99.90**	**99.80**	**100.00**	99.43	99.92	70.82	**100.00**	**99.99**	96.48±9.64
CDCN++ (Ours)	98.07	**99.90**	99.60	99.98	**99.89**	**99.98**	72.29	**100.00**	99.98	**96.63±9.15**

Table 7. The results of cross-dataset testing between CASIA-MFSD and Replay-Attack. The evaluation metric is HTER(%). The **multiple-frame** based methods are shown in the upper half part while **single-frame** based methods in bottom half part.

Method	Train CASIA-MFSD	Test Replay-Attack	Train Replay-Attack	Test CASIA-MFSD
Motion-Mag [5]	50.1		47.0	
LBP-TOP [16]	49.7		60.6	
STASN [62]	31.5		30.9	
Auxiliary [36]	27.6		28.4	
FAS-TD [56]	17.5		**24.0**	
LBP [7]	47.0		39.6	
Spectral cubes [48]	34.4		50.0	
Color Texture [8]	30.3		37.7	
FaceDs [26]	28.5		41.1	
CDCN (Ours)	15.5		32.6	
CDCN++ (Ours)	**6.5**		29.8	

Cross-type Testing. Following the protocol proposed in [3], we use CASIA-MFSD [66], Replay-Attack [13] and MSU-MFSD [57] to perform intra-dataset cross-type testing between replay and print attacks. As shown in Table 6, our proposed CDC based methods achieve the best overall performance (even outperforming the zero-shot learning based method DTN [37]), indicating our consistently good generalization ability among unknown attacks. Moreover, we also conduct the cross-type testing on the latest SiW-M [37] dataset and achieve the best average ACER (12.7%) and EER (11.9%) among 13 attacks. The detailed results are shown in ***Appendix C***.

Cross-dataset Testing. In this experiment, there are two cross-dataset testing protocols. One is that training on the CASIA-MFSD and testing on Replay-Attack, which is named as protocol CR; the second one is exchanging the training dataset and the testing dataset, named protocol RC. As shown in Table 7, our proposed CDCN++ has 6.5% HTER on protocol CR, outperforming the prior state-of-the-art by a convincing margin of 11%. For protocol RC, we also outperform state-of-the-art frame-level methods (see bottom half part of Table 7). The performance might be further boosted via introducing the similar temporal dynamic features in Auxiliary [36] and FAS-TD [56].

4.6. Analysis and Visualization.

In this subsection, two perspectives are provided to demonstrate the analysis why CDC performs well.

Robustness to Domain Shift. Protocol-1 of OULU-NPU is used to verify the robustness of CDC when encountering the domain shifting, i.e., huge illumination difference between train/development and test set. Fig. 7 shows that the network using vanilla convolution has low ACER on development set (blue curve) while high ACER on test set (gray curve), which indicates vanilla convolution is easily overfitted in seen domain but generalizes poorly when illumination changes. In contrast, the model with CDC is able to achieve more consistent performance on both development (red curve) and test set (yellow curve), indicating the robustness of CDC to domain shifting.

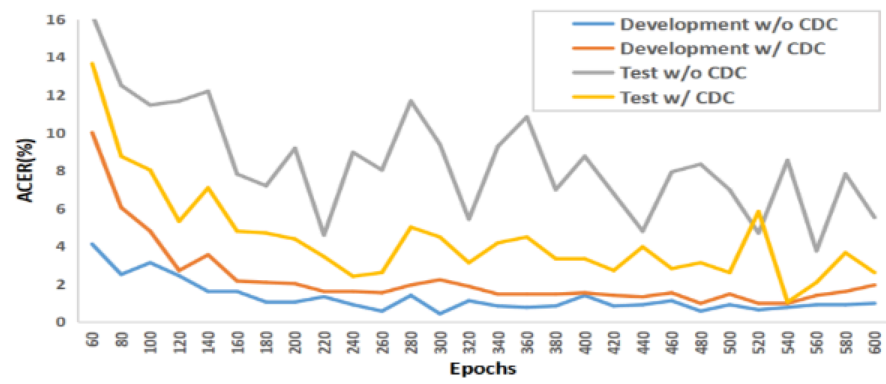

Figure 7. The performance of CDCN on development and test set when training on Protocol-1 OULU-NPU.

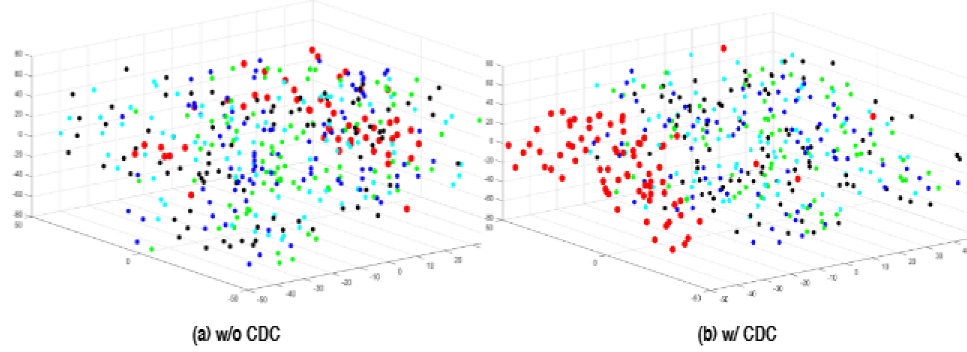

Figure 8. 3D visualization of feature distribution. (a) Features w/o CDC. (b) Features w/ CDC. Color code used: red=live, $green$=printer1, $blue$=printer2, $cyan$=replay1, $black$=replay2.

Features Visualization. Distribution of multi-level fused features for the testing videos on Protocol-1 OULU-NPU is shown in Fig. 8 via t-SNE [39]. It is clear that the features with CDC (Fig. 8(a)) presents more well-clustered behavior than that with vanilla convolution (Fig. 8(b)), which demonstrates the discrimination ability of CDC for distinguishing the living faces from spoofing faces. The visualization of the feature maps (w/o or w/ CDC) and attention maps of MAFM can be found in ***Appendix D***.

5. Conclusions and Future Work

In this paper, we propose a novel operator called Central Difference Convolution (CDC) for face anti-spoofing task. Based on CDC, a Central Difference Convolutional Network (CDCN) is designed. We also propose CDCN++, con-

表 6. 模型跨类型测试在 CASIA-MFSD、Replay-Attack 和 MSU-MFSD 上的 AUC (%)

方法	CASIA-MFSD [66] 视频	切割照片	包裹照片	Replay-Attack [13] 视频	数字照片	印刷照片	MSU-MFSD [57] 印刷照片	HR 视频	手机视频	总体
OC-SVM+BSIF [3]	70.74	60.73	95.90	84.03	88.14	73.66	64.81	87.44	74.69	78.68±11.74
SVM+LBP [10]	91.94	91.70	84.47	99.08	98.17	87.28	47.68	99.50	97.61	88.55±16.25
NN+LBP [59]	94.16	88.39	79.85	99.75	95.17	78.86	50.57	99.93	93.54	86.69±16.25
DTN [37]	90.0	97.3	97.5	99.9	99.9	99.6	81.6	99.9	97.5	95.9±6.2
CDCN（我们）	98.48	99.90	99.80	100.00	99.43	99.92	70.82	100.00	99.99	96.48±9.64
CDCN++（我们）	98.07	99.90	99.60	99.98	99.89	99.98	72.29	100.00	99.98	96.63±9.15

表 7. CASIAMFSD 和 Replay-Attack 之间的跨数据集测试结果。评估指标是 HTER(%)。基于多帧的方法显示在上半部分，而基于单帧的方法显示在下半部分。

方法	训练 CASIAMFSD / 测试 重放-攻击	训练 重放-攻击 / 测试 CASIA-MFSD
运动-磁 [5]	50.1	47.0
LBP-TOP [16]	49.7	60.6
STASN [62]	31.5	30.9
辅助 [36]	27.6	28.4
FAS-TD [56]	17.5	24.0
LBP [7]	47.0	39.6
光谱立方体 [48]	34.4	50.0
色彩纹理 [8]	30.3	37.7
FaceDs [26]	28.5	41.1
CDCN（我们）	15.5	32.6
CDCN++（我们）	6.5	29.8

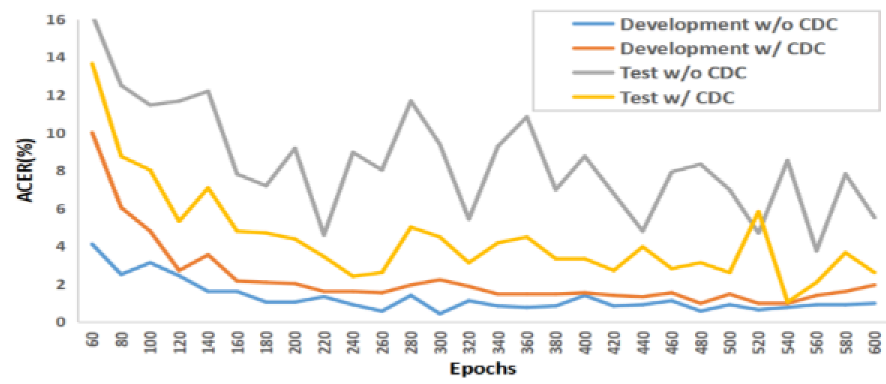

图 7. CDCN 在训练协议-1 OULU-NPU 时在开发和测试集上的性能表现。

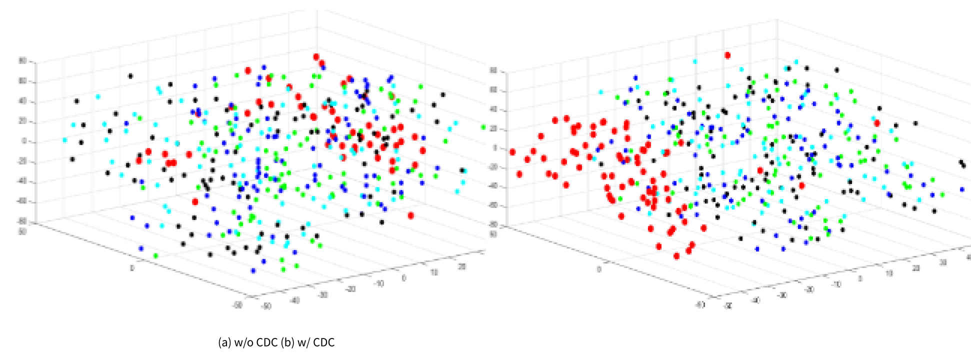

图 8. 特征分布的 3D 可视化。(a) 无 CDC 的特征。(b) 有 CDC 的特征。颜色代码使用：红色=活跃，绿色=打印机 1，蓝色=打印机 2，青色=重放 1，黑色=重放 2。

跨类型测试。遵循文献[3]中提出的协议，我们使用 CASIA-MFSD [66]、Replay-Attack [13]和 MSU-MFSD [57]在重放攻击和打印攻击之间进行数据集内跨类型测试。如表 6 所示，我们提出基于 CDC 的方法在整体性能上取得了最佳结果（甚至优于基于零样本学习的方法 DTN [37]），这表明我们在未知攻击中始终具有出色的泛化能力。此外，我们还对最新的 SiWM [37]数据集进行了跨类型测试，在 13 种攻击中实现了最佳的 ACER（12.7%）和 EER（11.9%）。详细结果见附录 C。

跨数据集测试。在这个实验中，有两种跨数据集测试协议。一种是使用 CASIA-MFSD 进行训练，在 Replay-Attack 上测试，称为协议 CR；另一种是交换训练数据集和测试数据集，称为协议 RC。如表 7 所示，我们提出的 CDCN++在协议 CR 上具有 6.5%的 HTER，比现有最佳方法高出 11%的显著优势。对于协议 RC，我们也优于现有的帧级方法（见表 7 的下半部分）。通过在辅助模块[36]和 FAS-TD[56]中引入类似的时序动态特征，性能可能得到进一步提升。

4.6. 分析与可视化。在本小节中，提供了两个视角来展示分析 CDC 表现优异的原因。

在域偏移方面，即训练/开发集和测试集之间存在巨大的光照差异。图 7 显示，使用普通卷积的网络在开发集（蓝色曲线）上 ACER 较低，而在测试集（灰色曲线）上 ACER 较高，这表明普通卷积在已知域中容易过拟合，但在光照变化时泛化能力较差。相比之下，带有 CDC 的模型在开发集（红色曲线）和测试集（黄色曲线）上都能实现更一致的性能，这表明 CDC 对域偏移的鲁棒性。

特征可视化。在协议-1 OULUNPU 上测试视频的多级融合特征分布通过 t-SNE [39]显示在图 8 中。很明显，带有 CDC（图 8(a)）的特征比带有普通卷积（图 8(b)）的特征表现出更好的聚类行为，这展示了 CDC 在区分真实面部和欺骗面部方面的辨别能力。特征图（无或带 CDC）和 MAFM 的注意力图的可视化可以在附录 D 中找到。

5. 结论与未来工作 在本文中，我们提出了一种称为中心差分卷积（CDC）的新算子用于面部防欺骗任务。基于 CDC，设计了一个中心差分卷积网络（CDCN）。我们还提出了 CDCN++，

领域偏移的鲁棒性。OULU-NPU 用于验证 CDC 在遇到

sisting of a searched CDC backbone and Multiscale Attention Fusion Module (MAFM). Extensive experiments are performed to verify the effectiveness of the proposed methods. We note that the study of CDC is still at an early stage. Future directions include: 1) designing context-aware adaptive CDC for each layer/channel; 2) exploring other properties (e.g., domain generalization) and applicability on other vision tasks (e.g., image quality assessment [34] and FaceForensics).

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由一个搜索的 CDC 主干和多尺度注意力融合模块（MAFM）组成。进行了大量实验以验证所提方法的有效性。我们注意到，CDC 的研究仍处于早期阶段。未来方向包括：1）为每一层/通道设计上下文感知的自适应 CDC；2）探索其他属性（例如，领域泛化）以及在其他视觉任务（例如，图像质量评估[34]和 FaceForensics）上的适用性。

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Appendix

A. Derivation and Code of CDC

Here we show the detailed derivation (Eq.(4) in draft) of CDC in Eq. (7) and Pytorch code of CDC in Fig. 9.

$$
\begin{aligned}
y(p_0) &= \underbrace{\theta \cdot \sum_{p_n \in \mathcal{R}} w(p_n) \cdot (x(p_0+p_n) - x(p_0))}_{\text{central difference convolution}} \\
&\quad + \underbrace{(1-\theta) \cdot \sum_{p_n \in \mathcal{R}} w(p_n) \cdot x(p_0+p_n)}_{\text{vanilla convolution}} \\
&= \underbrace{\theta \cdot \sum_{p_n \in \mathcal{R}} w(p_n) \cdot x(p_0+p_n)}_{\text{vanilla convolution}} + \underbrace{\theta \cdot (-\sum_{p_n \in \mathcal{R}} w(p_n) \cdot x(p_0))}_{\text{central difference term}} \\
&\quad + \underbrace{(1-\theta) \cdot \sum_{p_n \in \mathcal{R}} w(p_n) \cdot x(p_0+p_n)}_{\text{vanilla convolution}} \\
&= \underbrace{(\theta + 1 - \theta) \cdot \sum_{p_n \in \mathcal{R}} w(p_n) \cdot x(p_0+p_n)}_{\text{vanilla convolution}} + \underbrace{\theta \cdot (-x(p_0) \cdot \sum_{p_n \in \mathcal{R}} w(p_n))}_{\text{central difference term}} \\
&= \underbrace{\sum_{p_n \in \mathcal{R}} w(p_n) \cdot x(p_0+p_n)}_{\text{vanilla convolution}} + \underbrace{\theta \cdot (-x(p_0) \cdot \sum_{p_n \in \mathcal{R}} w(p_n))}_{\text{central difference term}}.
\end{aligned}
\tag{7}
$$

```
import torch.nn as nn
import torch.nn.functional as F
class CDC(nn.Module):
    def __init__(self, IC, OC, K=3, P=1, theta=0.7):
        # IC, OC: in_channels, out_channels
        # K, P: kernel_size, padding
        # theta: hyperparameter in CDC
        super(CDC, self).__init__()
        self.vani = nn.Conv2d(IC, OC, kernel_size=K, padding=P)
        self.theta = theta

    def forward(self, x):
        # x: input features with shape [N,C,H,W]
        out_vanilla = self.vani(x)

        kernel_diff = self.conv.weight.sum(2).sum(2)
        kernel_diff = kernel_diff[:, :, None, None]
        out_CD = F.conv2d(input=x, weight=kernel_diff, padding=0)

        return out_vanilla - self.theta * out_CD
```

Figure 9. Python code of CDC based on Pytorch.

B. Adaptive θ for CDC

Although the best hyperparameter $\theta = 0.7$ can be manually measured for face anti-spoofing task, it is still troublesome to find the best-suited θ when applying Central Difference Convolution (CDC) to other datasets/tasks. Here we treat θ as the data-driven learnable weights for each layer. A simple implementation is to utilize $Sigmoid(\theta)$ to guarantee the output range within $[0, 1]$.

As illustrated in Fig. 10(a), it is interesting to find that the values of learned weights in low (2nd to 4th layer) and high (8th to 10th layer) levels are relatively small while that in mid (5th to 7th layer) level are large. It indicates that the central difference gradient information might be more important for mid level features. In terms of the performance comparison, it can be seen from Fig. 10(b) that adaptive CDC achieves comparable results (1.8% vs. 1.0% ACER) with CDC using constant $\theta = 0.7$.

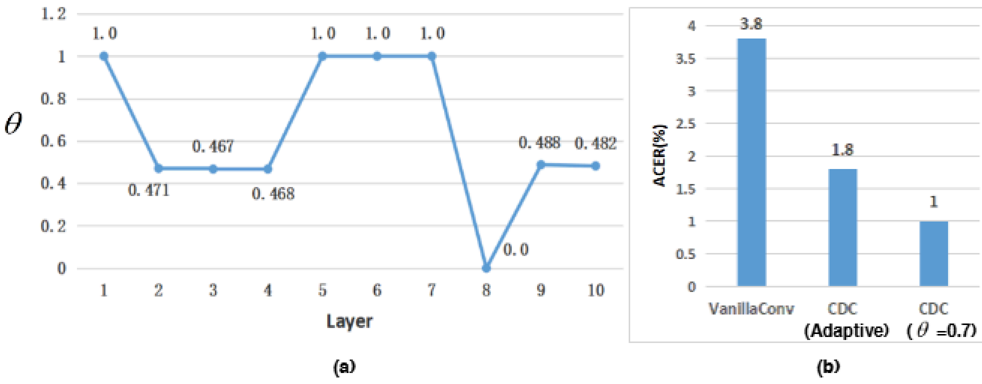

Figure 10. Adaptive CDC with learnable θ for each layer. (a) The learned θ weights for the first ten layers. (b) Performance comparison on Protocol-1 OULU-NPU.

C. Cross-type Testing on SiW-M

Following the same cross-type testing protocol (13 attacks leave-one-out) on SiW-M dataset [37], we compare our proposed methods with three recent face anti-spoofing methods [10, 36, 37] to valid the generalization capacity of unseen attacks. As shown in Table 8, our CDCN++ achieves an overall better ACER and EER, with the improvement of previous state-of-the-art [37] by 24% and 26% respectively. Specifically, we detect almost all "Impersonation" and "Partial Paper" attacks (EER=0%) while the previous methods perform poorly on "Impersonation" attack. It is obvious that we reduce the both the EER and ACER of Mask attacks ("HalfMask", "SiliconeMask", "TransparentMask" and "MannequinHead") sharply, which shows our CDC based methods generalize well on 3D non-planar attacks.

D. Feature Visualization

The low-level features and corresponding spatial attention maps of MAFM are visualized in Fig. 11. It is clear that both the features and attention maps between living and spoofing faces are quite different. 1) For the low-level features (see 2nd and 3rd row in Fig. 11), neural activation from the spoofing faces seems to be more homogeneous between the facial and background regions than that from living faces. It's worth noting that features with CDC are more likely to capture the detailed spoofing patterns (e.g., **lattice artifacts** in "Print1" and **reflection artifacts** in "Replay2"). 2) For the spatial attention maps of MAFM (see 4th row in Fig. 11), all the regions of hair, face and background have the relatively strong activation for the living faces while the facial regions contribute weakly for the spoofing faces.

附录

A. CDC 的推导和代码

这里我们展示了 CDC 的详细推导过程（草稿中的公式(4)）以及公式(7)中的 CDC 和图 9 中的 Pytorch 代码。

$$
\begin{aligned}
y(p) &= \theta \cdot \underbrace{\sum_{pn\in R} w(p) \cdot (x(p+p) - x(p))}_{\text{central difference convolution}} + (1-\theta) \cdot \underbrace{\sum_{pn\in R} w(p) \cdot x(p+p)}_{\text{vanilla convolution}} \\
&= \theta \cdot \underbrace{\sum_{pn\in R} w(p) \cdot x(p+p)}_{\text{vanilla convolution}} + \theta \cdot \underbrace{(-\sum_{pn\in R} w(p) \cdot x(p))}_{\text{central difference term}} + (1-\theta) \cdot \underbrace{\sum_{pn\in R} w(p) \cdot x(p+p)}_{\text{vanilla convolution}} \\
&= (\theta + 1 - \theta) \cdot \underbrace{\sum_{pn\in R} w(p) \cdot x(p+p)}_{\text{vanilla convolution}} + \theta \cdot \underbrace{(-x(p) \cdot \sum_{pn\in R} w(p))}_{\text{central difference term}} \\
&= \underbrace{\sum_{pn\in R} w(p) \cdot x(p+p)}_{\text{vanilla convolution}} + \theta \cdot \underbrace{(-x(p) \cdot \sum_{pn\in R} w(p))}_{\text{central difference term}} .
\end{aligned}
\tag{7}
$$

```
import torch.nn as nn
import torch.nn.functional as F
class CDC(nn.Module):
    def init(self, IC, OC, K=3, P=1, theta=0.7): # IC, OC: in channels, out
    channels # K, P: kernel size, padding # theta: hyperparameter in
    CDC
    super(CDC, self).__init__()
    self.vani = nn.Conv2d(IC, OC, kernel_size=K, padding=P)
    self.theta = theta


    def forward(self, x): # x: input features with shape
    [N,C,H,W]
        out_vanilla = self.vani(x)

       kernel diff = self.conv.weight.sum(2).sum(2) kernel diff = kernel
       diff[:, :, None, None] out CD = F.conv2d(input=x, weight=kernel
       diff, padding=0)

       return out vanilla − self.theta * out CD
```

图 9. 基于 Pytorch 的 CDC Python 代码。

B. CDC 的自适应θ

尽管对于人脸反欺骗任务，最佳超参数θ = 0.7 可以手动测量，但在将中心差分卷积（CDC）应用于其他数据集/任务时，寻找最合适的θ仍然很麻烦。在这里，我们将θ视为每个层的可学习的数据驱动权重。一个简单的实现方法是通过 Sigmoid(θ)来保证输出范围在[0, 1]内。

如图 10(a)所示，有趣的是发现，在低层（第 2 层至第 4 层）和高层（第 8 层至第 10 层）的权重值相对较小，而中层（第 5 层至第 7 层）的权重值较大。这表明

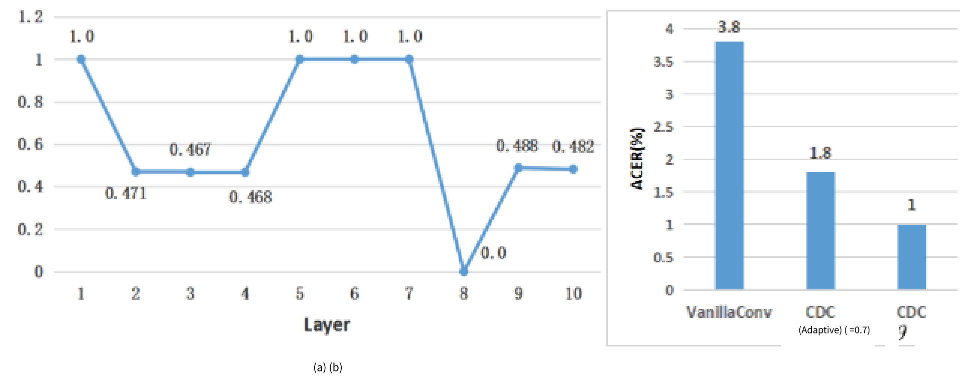

图 10. 每层具有可学习θ的自适应 CDC。(a) 前十层的可学习θ权重。(b) 在协议-1 OULU-NPU 上的性能比较。

中心差分梯度信息可能对中层特征更重要。在性能比较方面，从图 10(b)可以看出，自适应 CDC 与使用常数θ=0.7 的 CDC 相比，实现了相当的结果（1.8% vs. 1.0% ACER）。

C. 在 SiW-M 上的跨类型测试

在 SiW-M 数据集[37]上遵循相同的跨类型测试协议（13 次攻击留一法），我们将我们提出的方法与三种最近的面部反欺骗方法[10, 36, 37]进行比较，以验证对未见攻击的泛化能力。如表 8 所示，我们的 CDCN++在整体 ACER 和 EER 上取得了更好的结果，分别比之前的 SOTA[37]提高了 24%和 26%。具体来说，我们检测了几乎所有“模仿”和“部分纸张”攻击（EER=0%），而之前的方 法在“模仿”攻击上表现不佳。很明显，我们显著降低了面具攻击（“半面罩”、“硅胶面具”、“透明面具”和“假人头”）的 EER 和 ACER，这表明我们的基于 CDC 的方法在 3D 非平面攻击上泛化得很好。

D. 特征可视化

MAFM 的低层特征及其对应的空间注意力图在图 11 中进行了可视化。很明显，真实人脸和伪装人脸的特征及注意力图之间存在显著差异。1) 对于低层特征（见图 11 的第 2 行和第 3 行），伪装人脸的神经激活在面部和背景区域之间似乎比真实人脸更加均匀。值得注意的是，具有 CDC 特征更容易捕捉到详细的伪装模式（例如，“Print1”中的网格伪影和“Replay2”中的反射伪影）。2) 对于 MAFM 的空间注意力图（见图 11 的第 4 行），对于真实人脸，头发、面部和背景区域都有相对较强的激活，而对于伪装人脸，面部区域的激活较弱。

Table 8. The evaluation and comparison of the cross-type testing on SiW-M [37].

Method	Metrics(%)	Replay	Print	Mask Attacks					Makeup Attacks			Partial Attacks			Average
				Half	Silicone	Trans.	Paper	Manne.	Obfusc.	Imperson.	Cosmetic	Funny Eye	Paper Glasses	Partial Paper	
SVM$_{RBF}$+LBP [10]	APCER	19.1	15.4	40.8	20.3	70.3	0.0	4.6	96.9	35.3	11.3	53.3	58.5	0.6	32.8±29.8
	BPCER	22.1	21.5	21.9	21.4	20.7	23.1	22.9	21.7	12.5	22.2	18.4	20.0	22.9	21.0±2.9
	ACER	20.6	18.4	31.3	21.4	45.5	11.6	13.8	59.3	23.9	16.7	35.9	39.2	11.7	26.9±14.5
	EER	20.8	18.6	36.3	21.4	37.2	7.5	14.1	51.2	19.8	16.1	34.4	33.0	7.9	24.5±12.9
Auxiliary [36]	APCER	23.7	7.3	27.7	18.2	97.8	8.3	16.2	100.0	18.0	16.3	91.8	72.2	0.4	38.3±37.4
	BPCER	10.1	6.5	10.9	11.6	6.2	7.8	9.3	11.6	9.3	7.1	6.2	8.8	10.3	8.9± 2.0
	ACER	16.8	6.9	19.3	14.9	52.1	8.0	12.8	55.8	13.7	**11.7**	49.0	40.5	5.3	23.6±18.5
	EER	14.0	4.3	11.6	12.4	24.6	7.8	10.0	72.3	10.1	**9.4**	21.4	18.6	4.0	17.0±17.7
DTN [37]	APCER	1.0	0.0	0.7	24.5	58.6	0.5	3.8	73.2	13.2	12.4	17.0	17.0	0.2	17.1±23.3
	BPCER	18.6	11.9	29.3	12.8	13.4	8.5	23.0	11.5	9.6	16.0	21.5	22.6	16.8	16.6 ±6.2
	ACER	9.8	**6.0**	15.0	18.7	36.0	4.5	7.7	48.1	11.4	14.2	**19.3**	19.8	8.5	16.8 ±11.1
	EER	10.0	**2.1**	14.4	18.6	26.5	**5.7**	9.6	50.2	10.1	13.2	**19.8**	20.5	8.8	16.1± 12.2
CDCN (Ours)	APCER	8.2	6.9	8.3	7.4	20.5	5.9	5.0	43.5	1.6	14.0	24.5	18.3	1.2	12.7±11.7
	BPCER	9.3	8.5	13.9	10.9	21.0	3.1	7.0	45.0	2.3	16.2	26.4	20.9	5.4	14.6 ±11.7
	ACER	**8.7**	7.7	11.1	**9.1**	20.7	4.5	5.9	44.2	2.0	15.1	25.4	19.6	3.3	13.6 ±11.7
	EER	**8.2**	7.8	8.3	**7.4**	20.5	5.9	**5.0**	47.8	1.6	14.0	24.5	18.3	1.1	13.1± 12.6
CDCN++ (Ours)	APCER	9.2	6.0	4.2	7.4	18.2	0.0	5.0	39.1	0.0	14.0	23.3	14.3	0.0	10.8±11.2
	BPCER	12.4	8.5	14.0	13.2	19.4	7.0	6.2	45.0	1.6	14.0	24.8	20.9	3.9	14.6±11.4
	ACER	10.8	7.3	**9.1**	10.3	**18.8**	**3.5**	**5.6**	**42.1**	**0.8**	14.0	24.0	**17.6**	**1.9**	**12.7±11.2**
	EER	9.2	5.6	**4.2**	11.1	**19.3**	5.9	**5.0**	**43.5**	**0.0**	14.0	23.3	**14.3**	**0.0**	**11.9±11.8**

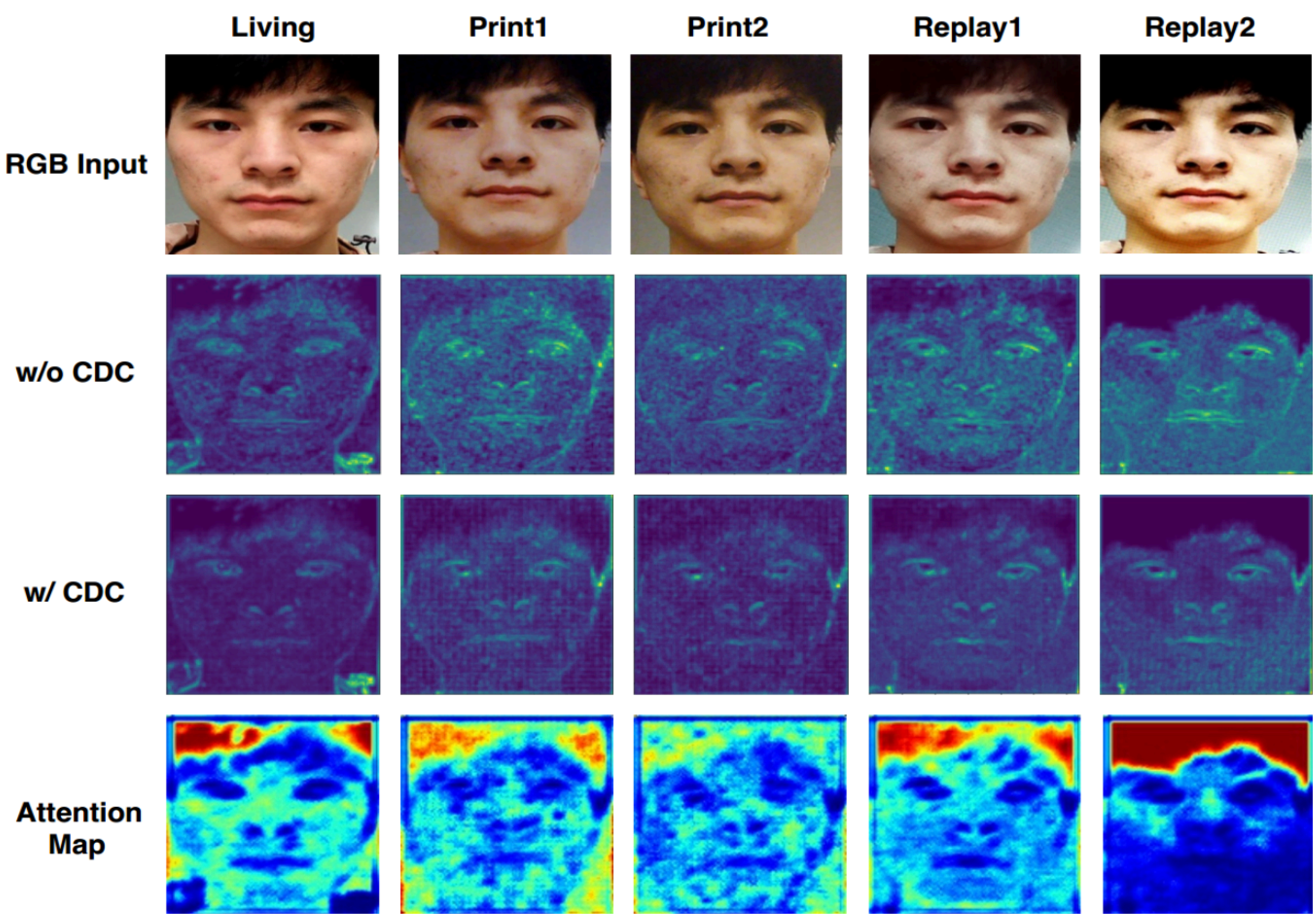

Figure 11. Features visualization on living face (the first column) and spoofing faces (four columns to the right). The four rows represent the RGB images, low-level features w/o CDC, w/ CDC and low-level spatial attention maps respectively. Best view when zoom in.

表 8. SiW-M [37]跨类型测试的评估与比较。

方法	指标(%)	重放	打印	掩码攻击					化妆攻击			部分攻击			平均
				半	硅胶	传输	纸张	人工模型	混淆	仿冒	化妆	滑稽眼睛	纸张眼镜	部分纸张	
SVM+LBP [10]	APCER	19.1	15.4	40.8	20.3	70.3	0.0	4.6	96.9	35.3	11.3	53.3	58.5	0.6	32.8±29.8
	BPCER	22.1	21.5	21.9	21.4	20.7	23.1	22.9	21.7	12.5	22.2	18.4	20.0	22.9	21.0±2.9
	ACER	20.6	18.4	31.3	21.4	45.5	11.6	13.8	59.3	23.9	16.7	35.9	39.2	11.7	26.9±14.5
	EER	20.8	18.6	36.3	21.4	37.2	7.5	14.1	51.2	19.8	16.1	34.4	33.0	7.9	24.5±12.9
辅助 [36]	APCER	23.7	7.3	27.7	18.2	97.8	8.3	16.2	100.0	18.0	16.3	91.8	72.2	0.4	38.3±37.4
	BPCER	10.1	6.5	10.9	11.6	6.2	7.8	9.3	11.6	9.3	7.1	6.2	8.8	10.3	8.9± 2.0
	ACER	16.8	6.9	19.3	14.9	52.1	8.0	12.8	55.8	13.7	11.7	49.0	40.5	5.3	23.6±18.5
	EER	14.0	4.3	11.6	12.4	24.6	7.8	10.0	72.3	10.1	9.4	21.4	18.6	4.0	17.0±17.7
DTN [37]	APCER	1.0	0.0	0.7	24.5	58.6	0.5	3.8	73.2	13.2	12.4	17.0	17.0	0.2	17.1±23.3
	BPCER	18.6	11.9	29.3	12.8	13.4	8.5	23.0	11.5	9.6	16.0	21.5	22.6	16.8	16.6 ±6.2
	ACER	9.8	6.0	15.0	18.7	36.0	4.5	7.7	48.1	11.4	14.2	19.3	19.8	8.5	16.8 ±11.1
	EER	10.0	2.1	14.4	18.6	26.5	5.7	9.6	50.2	10.1	13.2	19.8	20.5	8.8	16.1± 12.2
CDCN (Ours)	APCER	8.2	6.9	8.3	7.4	20.5	5.9	5.0	43.5	1.6	14.0	24.5	18.3	1.2	12.7±11.7
	BPCER	9.3	8.5	13.9	10.9	21.0	3.1	7.0	45.0	2.3	16.2	26.4	20.9	5.4	14.6 ±11.7
	ACER	8.7	7.7	11.1	9.1	20.7	4.5	5.9	44.2	2.0	15.1	25.4	19.6	3.3	13.6 ±11.7
	EER	8.2	7.8	8.3	7.4	20.5	5.9	5.0	47.8	1.6	14.0	24.5	18.3	1.1	13.1± 12.6
CDCN++ (Ours)	APCER	9.2	6.0	4.2	7.4	18.2	0.0	5.0	39.1	0.0	14.0	23.3	14.3	0.0	10.8±11.2
	BPCER	12.4	8.5	14.0	13.2	19.4	7.0	6.2	45.0	1.6	14.0	24.8	20.9	3.9	14.6±11.4
	ACER	10.8	7.3	9.1	10.3	18.8	3.5	5.6	42.1	0.8	14.0	24.0	17.6	1.9	12.7±11.2
	EER	9.2	5.6	4.2	11.1	19.3	5.9	5.0	43.5	0.0	14.0	23.3	14.3	0.0	11.9±11.8

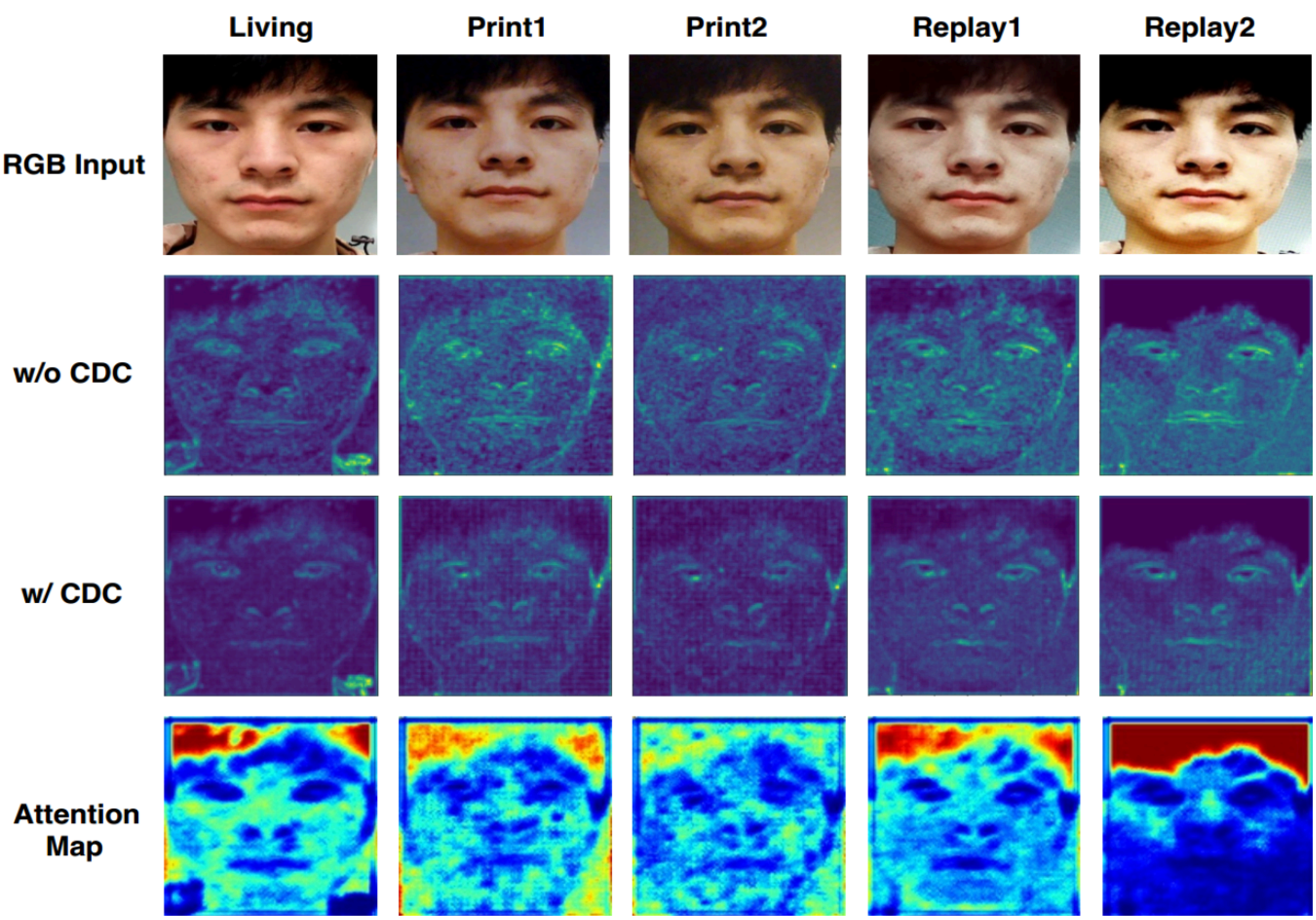

图 11. 活体人脸（第一列）和伪造人脸（右侧四列）的特征可视化。四行分别代表 RGB 图像、无 CDC 的低级特征、带 CDC 的低级特征和低级空间注意力图。最佳观看效果为放大查看。